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A Coupled Chemo-Mechanical and Life-Cycle Framework for Oyster-Shell-Derived Cementitious Materials in Aggressive Infrastructure Environments

Leon Sandler[1],*

[1] Independent Researcher, Northbrook, Illinois, United States * Corresponding author: sandler.leon@gmail.com ORCID: 0009-0007-4584-808X

Abstract

Oyster-shell powder is promoted as a low-carbon cement substitute, but joint carbon and durability benefits are usually asserted rather than tested. This paper asks when shell-derived concrete (shell replacing cement) delivers a carbon cut with non-inferior service life relative to ordinary Portland cement (OPC) concrete. A coupled hydration–transport–damage–carbon model is run through paired Monte Carlo analysis of a pavement, a dam and a marine culvert. Published data reject the original inert-filler strength model. An effective strength-efficiency coefficient k_f calibrated on one scallop-shell dataset does not generalize to independent oyster-shell data (mean absolute error 0.12 against 0.15 for the inert model; root-mean-square error 0.17 against 0.15), and k_f ranges from -0.40 to 1.36 across four studies, so the primary results use a pooled random-effects distribution. At 25% replacement and constant water-to-binder ratio the median carbon change is −21% with strength 0.88 of OPC; with strength matched it is −15% (P10–P90 −22 to +10%). In the primary configuration (pooled strength-only k_f, sampled diffusivity mapping, OPC reference) the probability of a carbon cut of at least 5% with non-inferior life is at most 0.21 for the pavement, 0.14 for the dam and 0.03 for the marine culvert, and zero against a 50% slag blended cement; other, equally unmeasured transport assumptions give 0.18 to 0.68 for the pavement. All probabilities are conditional scenario results, not empirical probabilities for real infrastructure, and the study is a falsifiable computational screening framework, not a validated service-life model.

Keywords: oyster shell; calcium carbonate filler; service-life prediction; embodied carbon; approximate Bayesian calibration; uncertainty and sensitivity analysis

1. Introduction

Cement production is a major source of anthropogenic CO_2, with roughly half of the process emissions arising from the calcination of limestone rather than from fuel (Andrew 2019). Strategies to reduce this footprint include clinker substitution by supplementary materials and fillers, alternative binders and carbon capture (Scrivener et al. 2018). Aquaculture waste shells, composed mainly of calcium carbonate, are an attractive candidate because they are a locally available, otherwise problematic waste. Technical communications on “oyster-shell green concrete” frequently claim large carbon reductions, fire resistance to very high temperatures, accelerated hydration, reduced shrinkage and superior chemical resistance. These claims are presented without a mathematical model, uncertainty analysis or experimental evidence, and they conflate materials pathways that are chemically and mechanically different.

1.1 What existing research establishes

Shell-derived calcium carbonate. Ground shells are predominantly calcium carbonate with a small organic matrix and trace salts. Experimental studies of ground seashells (including oyster shell) as partial cement replacement in mortars and masonry blocks report that compressive strength decreases as the replacement fraction increases, that water absorption increases in blocks, that setting is retarded, and that low replacement levels (about 10%) can still meet standards, with higher levels reaching required strength only at later ages (Lertwattanaruk et al. 2012; Jasni et al. 2024). Drying shrinkage of mortars containing some shell types was reported to decrease (Lertwattanaruk et al. 2012). These studies do not provide the kinetic, transport and damage parameters needed for a predictive treatment, and they do not address carbon and life together. A recent constant-binder study of scallop-shell powder reports about 25% lower 28-day strength at 40% binder replacement and a lower carbon footprint (Abbas and Kudukkan 2025); it is the out-of-sample dataset used in Section 4.4.

Hydration and the filler effect. Fine calcium carbonate accelerates early cement hydration through heterogeneous nucleation of C–S–H and, in the presence of aluminates, forms carboaluminates that stabilise ettringite (Lothenbach et al. 2008; Berodier and Scrivener 2014). Dilution, however, raises the effective water-to-cement ratio and removes reactive material, so strength generally decreases with replacement. Organic matter associated with shells may retard hydration.

Transport and durability. Chloride ingress is commonly modeled by Fick's second law with an age-dependent diffusivity; the Life-365 formulation relates the 28-day diffusivity of OPC concrete to the water-to-cementitious ratio and uses an age exponent of 0.2 (Thomas and Bentz 2002). Acid attack and calcium leaching are classically described by moving-front models in which front depth grows with the square root of time and decreases with the acid-neutralising or calcium capacity of the matrix. Calcite can neutralise acid but also dissolves, so its net effect depends on exposure.

Damage and service life. Fatigue of plain concrete follows stress–life (S–N) relations such as that of Aas-Jakobsen (1970); freeze–thaw damage is governed by the degree of saturation relative to a critical value (Fagerlund 1977); reinforcement corrosion is treated with an initiation plus propagation framework (Tuutti 1982); and continuum damage mechanics treats degradation through an effective stress $\sigma/(1 - D)$ (Kachanov 1958; Lemaitre 1996).

Life-cycle assessment. Cradle-to-gate embodied carbon is dominated by the cement. Shell powder replaces cement at a small fraction of its footprint, but processing (washing, drying, grinding) and transport are non-zero, and for P2 the calcination CO_2 per unit $CaCO_3$ is stoichiometrically independent of the carbonate source.

A competing use of shell. Shell is also a restoration resource. It is a widely used substrate for oyster-reef restoration, its supply can limit restoration (Brumbaugh and Coen 2009; Petrolia and Mohrman 2025), and reefs provide habitat, filtration and shoreline services (Beck et al. 2011; Zu Ermgassen et al. 2012; Grabowski et al. 2012). Shell used in concrete is therefore not free of an opportunity cost. That trade-off is the subject of a companion paper on the allocation of recovered shell between reef restoration and construction (Sandler 2026b); here it is labelled pathway P3 and used as context only, and this paper is restricted to the concrete pathways P1 and P2. The companion paper takes results from this one as inputs; it is companion work and not independent validation of them.

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1.2 Research gap, claims and hypotheses

A title-level screen of the Crossref index on 1 November 2026 (terms combining oyster or seashell with cement replacement, strength, service life, chloride, life-cycle carbon and uncertainty) returned experimental studies and reviews of shell as a cement replacement but, to the author's knowledge, no model that couples processing-dependent carbon, hydration, chloride and chemical transport, damage accumulation and infrastructure-specific service life under shared parametric uncertainty. This is a limited screen and not a systematic review, so the gap is stated conditionally. The source technical white paper made seven performance claims. Rather than accept them, each is converted into a testable hypothesis (Table 1).

Table 1. Claims in the source white paper and how each is treated in this manuscript.

Claim	Treatment here
Up to 80% CO_2 reduction	Cradle-to-gate carbon model with explicit processing, transport, calcination and credit terms; carbon change reported at constant w/b and at constant strength
Thermal resistance to 2,400 °F	Not modeled; assessed against calcite thermochemistry only (Section 8)
100% limestone substitution	Separated: P2 (feedstock replacement) vs P1 (cement replacement); never placed on one axis
Accelerated hydration, reduced shrinkage	Hydration kinetics with explicit filler acceleration and organic retardation; shrinkage and early-age cracking indices
Superior acid and chemical resistance	Moving-front acid and leaching models in which calcite both buffers and dilutes
Higher structural durability	Service life from damage accumulation and chloride corrosion limit states, paired with an OPC reference
Inorganic-polymer (alkali-activated) bonding	Excluded (pathway P4); shell calcium carbonate is not an aluminosilicate precursor

Research question. Under which processing, substitution and exposure conditions does oyster-shell-derived concrete provide a simultaneous carbon and durability advantage over conventional concrete, and under which does it fail? Formally, we define a falsification region (Section 3.5) and estimate its probability under parameter uncertainty. The hypotheses are: H1, shell replacement at constant w/b lowers carbon but not service life; H2, at constant strength the carbon benefit shrinks and is lost wherever shell contributes little to strength; H3, the P2 pathway offers no carbon advantage unless calcination CO_2 is credited; H4, shell calcite improves acid resistance only when its particles are fine enough to be accessible.

2. Materials and processing pathways

Oyster shell is approximately 95–98% calcium carbonate (predominantly calcite in the foliated layers) with 1–4 wt% organic matrix (conchiolin) and trace sodium, magnesium, chloride, phosphorus and sulfur; these ranges are generic and should be replaced by measurements for a given supply. Processing for cementitious use comprises collection, weathering or washing to reduce organics and salt, drying, crushing and fine grinding to a target median particle size (10–200 μm is explored). Four pathways are distinguished (Table 2).

Table 2. Pathways considered. They are not interchangeable and are modeled on separate axes.

Pathway	Material route	Principal mechanisms	Status here
P1	Uncalcined shell powder replaces a mass fraction r of cement in concrete	Dilution, filler nucleation, packing, organic retardation, calcite acid buffering	Modelled; r = 10–100%, median size 10–200 µm
P2	Shell $CaCO_3$ replaces a fraction ρ of limestone in clinker raw meal; cement chemistry unchanged	Impurity effects on cement; processing and transport carbon; calcination CO_2 unchanged per kg $CaCO_3$	Modelled; ρ = 25–100%, biogenic credit 0–100%
P3	Recovered, cleaned shell returned to the water as reef substrate (competing use)	Recruitment substrate; habitat, filtration and shoreline services; transport and cleaning energy	Context only; allocation modeled in the companion paper (Sandler 2026b)
P4	Shell-based alkali-activated binders	Requires an aluminosilicate precursor; not a calcium-carbonate substitution	Out of scope

3. Theoretical framework

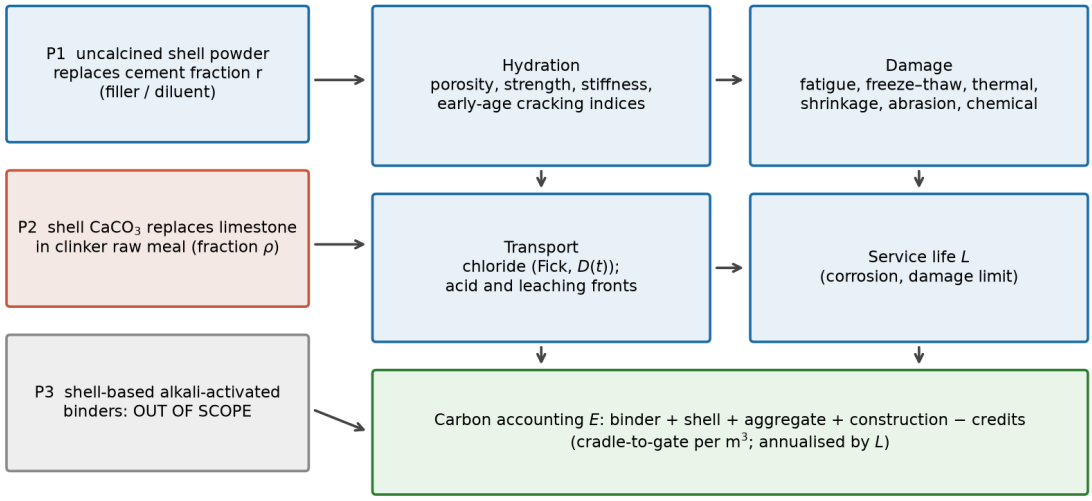


Fig. 1. Coupled model architecture. Pathways P1 and P2 feed hydration, which sets porosity, strength and stiffness; these drive transport and damage, which determine service life; carbon accounting is evaluated per m^3 and per year of life.

All quantities are in SI-consistent units; Appendix B lists symbols and units. Damage D, stress ratio S and degree of saturation are dimensionless, and every rate in the damage law has units of inverse time. All parameters are uncertain and are sampled (Section 4). This is a **proposed phenomenological framework**, not a validated constitutive model; each component is anchored to an established relation where one exists and its limits are stated.

3.1 Hydration, porosity and strength

Hydration degree α evolves with a reaction-progress law that can be accelerated or retarded by shell addition:

$$d\alpha/dt = k_0\, A(T)\, (1 + \nu\, s(d)\, r)\, f_{ret}\, (\alpha_{max} - \alpha)^n \quad (1)$$

where k_0 is the rate constant at 20 °C, $A(T) = \exp[E_a/R\,(1/293.15 - 1/T)]$ is the Arrhenius factor, ν is the filler nucleation coefficient, $s(d) = \min[(30\ \mu m/d)^{0.5}, 2]$ is a fineness factor, $f_{ret} = 1/(1 + \kappa_o\, q)$ is the retardation factor with $q = o(1 - \eta_w)\, r/(1 - r)$ the residual organic dose per unit cement (o is the organic

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fraction and η_w the washing efficiency), and α_max = min[α_cap, (w/c)/0.42] limits the attainable degree to the water available per unit cement. For α(0) = 0 and n ≠ 1 the closed form is

$$\alpha(t) = \alpha_{max} - [\alpha_{max}^{(1-n)} + (n-1)\, k\, t]^{(1/(1-n))} \quad (2)$$

Shell addition therefore raises the rate through ν and the ceiling through a larger effective w/c, but lowers the rate through organics, and removes cement. With cement volume v_c, water volume v_w, hydration products of volume 2.06 α v_c, the capillary porosity is φ_cap = v_w − 1.06 α v_c (Powers and Brownyard 1947) and the gel/space ratio X = 2.06 α v_c/(v_w + α v_c). Compressive strength is

$$f_c = S_0' X^{\kappa} [1 + \beta_p\, s\, r(1-r)] (1 - k_{imp}\, \rho) \quad (3)$$

where the exponent κ is uncertain (1.8–3.0) and $S_0' = S_0 \times 0.742^{(2-\kappa)}$ keeps the nominal OPC strength at w/c = 0.45 fixed for any κ; β_p is a packing bonus and k_imp an impurity penalty that applies only to P2. Stiffness, tensile and flexural strength follow $E = 4700\sqrt{f_c}$, $f_t = 0.33\sqrt{f_c}$ and $f_r = 0.62\sqrt{f_c}$ (MPa; ACI 318).

Effective strength-efficiency coefficient. In the original model shell powder is an inert diluent in the strength relation. To test this against published data (Section 4.4) an optional coefficient k_f is introduced: the cement volume entering the gel/space ratio is multiplied by [1 + k_f r/(1 − r)], so that a unit mass of shell counts as k_f units of cement for strength, as in the classical k-value concept for fillers and supplementary materials. k_f is an empirical lump of packing, nucleation, organic retardation, particle-size and other unresolved effects; it is not the chemical binding capacity of shell and a value of 0.65 does not mean that shell has 65% of the binding capacity of cement. Hydration, porosity and transport are not changed in the primary formulation; two alternatives (Section 7.4) couple k_f to the porosity or make it depend on particle size. k_f = 0 recovers the original model.

3.2 Transport and chemical degradation

Chloride. One-dimensional diffusion with spatially uniform, age-dependent diffusivity has the solution $C(x,t) = C_s\, \mathrm{erfc}[x/(2\sqrt{\tau})]$ with $\tau = \int D\, dt$ (Crank 1975). The diffusivity is

$$D(t) = D_{28}\, f_T\, (28\ \mathrm{d} / t_{age})^{m} [1 + \kappa_c\, D_{mech}(t)], \quad m = m_0 + \omega\, r \quad (4)$$

where f_T is an Arrhenius temperature factor (activation energy 35–50 kJ/mol; Life-365 uses 44.6 kJ/mol), $m_0 \approx 0.2$ (Thomas and Bentz 2002), ω an uncertain filler shift and κ_c couples mechanical damage to transport (cracks raise diffusivity). Because the damage trajectory does not depend on chloride, the coupling is exact for constant surface chloride. The 28-day diffusivity is a blend of two mappings with an uncertain weight χ_D ~ U(0,1), which represents **model-form uncertainty**:

$$\ln D_{28} = (1 - \chi_D) \ln D_\varphi + \chi_D \ln D_{wc} \quad (5)$$

$D_\varphi = \lambda_D\, D_{ref}\, (\varphi_p/\varphi_{ref})^{(\zeta + \Delta\zeta)}$ is a porosity mapping, with the paste capillary porosity φ_p and the exponent ζ = 1.73 fitted to the Life-365 relation $D_{28} = 10^{(-12.06 + 2.40\ \mathrm{w/cm})}$ m²/s for OPC over w/c = 0.30–0.60 (model-to-Life-365 ratio within 1.23); D_wc applies the same Life-365 relation to the cement-only water-to-cement ratio, which treats filler as an inert diluent (pessimistic). Applying a mapping calibrated on OPC to filler-bearing pastes is an extrapolation, which is why both are retained. Corrosion initiates when C(x_cover, t) reaches a critical chloride content and a propagation period follows (Tuutti 1982).

Acid and leaching fronts. For a sharp reaction front, $x\,dx/dt = D_e\,C_a/N$ gives $x = \sqrt{(2 \int D_e\,dt\,C_a/N)}$, where C_a is the aggressive-species concentration and N the capacity of the matrix per unit volume. For acid attack, $N = 2[\xi_N \cdot 11.6\,\alpha\,c + \chi(d)\,m_{calcite}/0.10009]$ (mol H^+/m^3), where cement supplies 11.6 mol Ca per kg, ξ_N is the accessible share, and calcite supplies 2 mol H^+ per mole but only when particles are fine enough: $\chi(d) = 1/[1 + (d/d_c)^2]$. For soft-water leaching, $N = \xi_L \cdot 11.6\,\alpha\,c$ (calcite does not contribute because of its low solubility), and a pressure-driven term $x_a = K\,i\,C_{sat}\,t/N$ is added in quadrature to the diffusive front, with hydraulic conductivity $K = \kappa_K\,D_e$. Mass loss from dissolved calcite is not tracked.

3.3 Damage accumulation

Total damage is the sum of a mechanical state D_m integrated in time and chemical and abrasion terms that are explicit functions of time:

$$dD_m/dt = n_{traf} / N_f(S_{eff}) + n_{ft}\,d_{ft}(S_w) \tag{6}$$

$$D(t) = D_0 + D_m(t) + D_f\,x_{chem}(t)/x_{lim} + D_f\,w_{abr}(t)/w_{lim} \tag{7}$$

Fatigue life follows the Aas-Jakobsen relation $\log_{10}N_f = (1 - S)/[0.0685(1 - R)]$ with stress ratio $R = 0.1$, and the effective stress ratio $S_{eff} = (\sigma_{traf} + \sigma_{env})/[f_r(t)(1 - D)]$ includes the continuum-damage amplification $1/(1 - D)$, so that damage accelerates as the section degrades. The traffic stress σ_{traf} is a fixed fraction (low, medium, high stress ratio class) of the OPC reference flexural strength, so that shell and OPC mixes carry the same absolute load; σ_{env} is the restrained shrinkage/curling stress. Freeze–thaw damage per cycle is $d_{ft} = d_1[(S_w - S_{crit})/(1 - S_{crit})]^q$ for $S_w > S_{crit}$, with $S_w = (\varphi_{cap} + a_f\,a_{air})/(\varphi_{cap} + a_{air})$ the degree of saturation (Fagerlund 1977). The chemical term has the form $D_f\,x/x_{lim}$, i.e. the failure damage D_f is reached exactly when the front reaches its limit, which fixes the coefficient B of a generic chemical driver by dimensional consistency rather than leaving it free. Early-age damage D_0 is switched on by a logistic function of the cracking indices $I_{sh} = \max(\sigma_{sh}/f_t)$ for restrained shrinkage (pavement) or I_{th} for restrained thermal contraction (dam); a mass-concrete temperature drop is $\Delta T = \psi_T\,c\,Q_{cem}\,\alpha(5\,d)/(\rho\,c_p)$.

Service life is the earliest of: chloride initiation plus propagation, and D reaching D_f; it is censored at 100 years. Because OPC reference concrete often survives the horizon, non-inferiority is defined with a tie-break: when the OPC reference survives 100 years, the shell mix must also survive and its damage index at 100 years must not exceed the reference by more than 5% of D_f.

3.4 Carbon accounting

$$E_{total} = E_{binder} + E_{shell} + E_{aggregate} + E_{water/admix} + E_{construction} - E_{credits} \tag{8}$$

The functional unit is 1 m^3 of concrete in service; cradle-to-gate embodied carbon E_{gate} excludes construction, and the annualized value divides E_{total} by service life. Cement carries a footprint EF_{cem} (0.75–0.95 kg CO_2e/kg). For P1, shell powder carries $EF_{sh} = [e_{dry}\,EF_{th} + E_g(d)\,EF_{el}]/1000 + (distance \times EF_{truck})/1000 - c_{land}$ per kg, where grinding energy scales as $E_g = E_{ref}\,(45\,\mu m/d)^{0.5}$ and c_{land} is a small landfill avoidance credit. For P2 the change in cement footprint per kg is $\Delta EF = \rho\,m_{CaCO_3}\,[EF_{proc} - EF_{lq} + EF_{kiln}] - \beta_{bio}\,\rho\,m_{CaCO_3} \times 0.4397$, where 0.4397 kg CO_2/kg $CaCO_3$ is the stoichiometric calcination emission and β_{bio} is the fraction of that emission credited as biogenic. **The base case sets $\beta_{bio} = 0$**: shell calcium carbonate does not eliminate the CO_2 released on calcination. Organic CO_2 is treated as biogenic.

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3.5 Falsification region

For uncertain parameters θ, let E be embodied carbon and L service life. The shell alternative fails to deliver a simultaneous advantage over its OPC reference, on the same draw of θ, in the set

$$\mathscr{F} = \{ (r, \theta) : E_shell \geq E_OPC \ \vee \ L_shell \text{ not non-inferior to } L_OPC \} \quad (9)$$

The probability $P(\mathscr{F})$ is estimated by Monte Carlo and its complement P_adv is reported. Because carbon differences of a fraction of a percent are within the noise of the inputs, a second, material version requires $E_shell \leq 0.95\ E_OPC$ and is reported as P_adv,mat. Both are given; the falsification maps use the material version.

4. Numerical methodology

4.1 Cases, parameters and sampling

Three cases are modeled with case-specific mix designs, exposures and mechanisms (Appendix A): (A) highway rigid pavement (fatigue, shrinkage, freeze–thaw, chloride from de-icing salt), (B) dam mass concrete (early-age thermal cracking, soft-water leaching with hydrostatic gradient, freeze–thaw), and (C) marine culvert / riverbed (chloride, acid, abrasion). The parameter study varies shell replacement (r = 10, 25, 50, 75, 100%), median particle size (10, 30, 100, 200 μm), chemical exposure (mild, moderate, severe), mechanical stress ratio (low, medium, high), relative humidity (55–100%), temperature (5–60 °C curing in the hydration study) and horizon (1–100 yr). The P2 pathway is varied over ρ = 25–100% and β_bio = 0, 50, 100%. For P1, two design rules are compared: **same w/b** (shell replaces cement at constant water and binder content, as in the source white paper) and **strength-matched** (binder content is increased at constant water until the 28-day strength equals the OPC reference; the search is restricted to the range in which the model strength increases with binder, and draws where matching is infeasible are flagged; 81% of draws are matchable at r = 25%).

Uncertain inputs (65 common and up to 11 case-specific parameters) are drawn by Latin hypercube sampling (McKay et al. 1979) with n = 3000. Every shell configuration is evaluated on the same draws as its OPC reference (common random numbers), so differences are attributable to the material and not to sampling noise. Global sensitivity uses Sobol total-order indices with the Jansen estimator (Saltelli et al. 2010; Sobol 2001), n = 2048 base samples with bootstrap intervals. The cases are time-stepped at 0.125 yr; hydration, chloride and front solutions are closed form or cumulative integrals.

4.2 Baseline anchoring and calibration

Three calibration steps use only the OPC reference. (i) The porosity–diffusivity exponent is fitted to the Life-365 relation (ζ = 1.73; maximum deviation factor 1.23). (ii) The strength model is set so that nominal OPC strength at w/c = 0.45 is about 43 MPa and the strength–w/c slope resembles an Abrams-type curve. (iii) Traffic cycles, curling stress, stress-ratio classes, surface chloride and cover are chosen so that the OPC reference reaches plausible design lives (median 24.9 yr for the pavement, 18.1 yr for the culvert, 98% survival to 100 yr for the dam). The last step anchors the absolute scale of the service life, so results should be read as **relative to the reference**, not as absolute life predictions. Shell-specific parameters are not calibrated in these three steps; they are tested and calibrated against published data in Section 4.4.

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4.3 Verification and validation

Table 3. Numerical verification (python ShellConcreteMC_v2.py --verify).

Test	Result
Baseline recovery: r = 0 (P1) and ρ = 0 (P2) reproduce OPC life and carbon	max difference 0
Closed-form hydration vs fourth-order Runge–Kutta integration	max error 4.5e-6
Time-varying-D chloride solution vs Crank–Nicolson finite differences	max error 2.2e-5 of C_s
Damage time-step convergence (Δt = 0.125 vs 0.0625 yr)	median 0.2%; 95th percentile 3.0% difference in life
Carbon budget closure (sum of parts = total)	residual 0
Calcination stoichiometry	0.4397 kg CO_2/kg $CaCO_3$
Strength matching reproduces the OPC 28-day strength (matched draws)	max relative error 1.3e-11; 81% feasible at r = 25%
D_{28} vs Life-365 relation, OPC, w/c = 0.30–0.60	within a factor 1.23

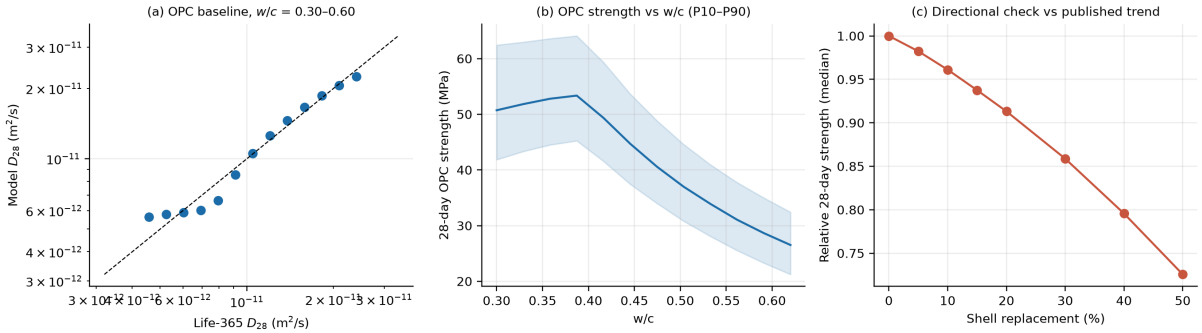


Fig. 2. Baseline checks. (a) Model D_{28} against the Life-365 relation for OPC. (b) Model OPC 28-day strength against w/c (P10–P90). (c) Median relative 28-day strength against shell replacement (P1, 25 µm).

Verification is not validation. The checks above show that the equations are solved correctly and that the baseline is recovered; they say nothing about whether the model describes shell concrete. The model is compared with published directional behavior (Table 4) and, in Section 4.4, with the only open quantitative dataset found. Conformity with directional trends is a necessary, not sufficient, condition.

Table 4. Directional checks of the model against published trends for ground shell as cement replacement.

Published trend	Model check	Result
Strength decreases monotonically with replacement (5–50%)	P(strength decreases at every step), r = 0–50%	60%
Strength relative to OPC at 10/20/50% replacement	Median relative 28-day strength	0.96/0.91/0.73
Porosity/water absorption increases with replacement	P(capillary porosity at 50% > at 0%)	100%
Drying shrinkage decreases (some shell types)	P(shrinkage proxy at 20% < at 0%)	100%
Setting/early hydration retarded by shell organics	P(α(1 d) lower; unwashed shell, no filler effect)	100%

4.4 Out-of-sample tests, calibration and the strength–efficiency coefficient

4.4.1 Calibration on a scallop-shell dataset

Summary of the validation. The strength model does not reliably generalize across shell materials. Calibrated on one scallop-shell study, it has a mean absolute error of 0.12 on 11 independent oyster-shell

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points, only slightly below the inert model (0.15) and a dilution-only rule (0.14), and a larger root-mean-square error than the inert model (0.17 against 0.15); it over-predicts the strength of the one study with a large replacement by about 0.37. No durability measurement on shell concrete enters the framework at all. The results below are therefore conditional scenario results, and the model must not be read as predicting the strength or the durability of real concrete or infrastructure.

Validation needs measured data. The first dataset used is Abbas and Kudukkan (2025), who used scallop shell as filler and partial cement replacement at a binder content of 350 kg/m³ and a water content of 160 kg/m³. The three mixes contain 50, 95 and 140 kg/m³ of shell-derived powder (r = 14%, 27% and 40% of the binder) with the median particle sizes reported there. Observed 28-day strengths relative to the control are read from the text for two mixes ("almost the same", "nearly 25% reduction") and derived for the third from the reported moduli of elasticity ($E \propto \sqrt{f_c}$); they are approximate, and a tolerance of ±0.07 on the relative strength represents this and specimen scatter.

Table 5. Calibration set: strength model against Abbas and Kudukkan (2025), 28-day compressive strength relative to the control mix, median (P10–P90) of model draws. Observed values: S0F15 and S30F15 from the text of the source, S15F15 derived from the reported moduli. The strength model is shared by all three cases.

Mix	Shell powder, r	Median size (µm)	Observed	Original model ($k_f = 0$)	With k_f prior	Calibrated (posterior)	Role
S0F15	14%	90	1.00	0.79 (0.75–0.83)	0.92 (0.81–1.03)	0.93 (0.93–0.94)	calibration
S15F15	27%	64	0.86	0.61 (0.54–0.68)	0.85 (0.65–1.05)	0.87 (0.87–0.88)	held out
S30F15	40%	55	0.75	0.43 (0.36–0.52)	0.77 (0.49–1.07)	0.81 (0.79–0.82)	calibration

The original model predicts relative strengths of 0.79, 0.61 and 0.43 against 1.00, 0.86 and 0.75 observed; none of 100,000 prior draws falls within the tolerance for any mix, so the inert-filler strength model is rejected by these data. Adding k_f (prior U(0, 1.2)) and running rejection approximate Bayesian computation on S0F15 and S30F15 (300,000 draws, tolerance 0.07) accepted 10,999 draws (3.7%); S15F15 was held out. The posterior median of k_f is 0.65 (P10–P90 0.54–0.84), and the posterior predictive for the held-out mix is 0.87–0.88 against 0.86 observed (Fig. 3a). This is an interpolation between the calibration mixes, so it is a weak test, and a bias of 0.06–0.07 remains at the calibration mixes (0.93 against 1.00 and 0.81 against 0.75), indicating a residual structural misfit of the strength relation. The carbon accounting was also checked against the same paper: with its emission factors (cement 0.91, shell powder 0.059 kg CO_2e/kg) the model arithmetic gives −24.4% and −36.0%, the same budget the model uses (a consistency check, not an independent test), and the model's own shell emission factor at 55 µm (0.058 kg CO_2e/kg, P10–P90 0.036–0.084) is of the same order as the laboratory value of 0.059 whose boundary is not stated.

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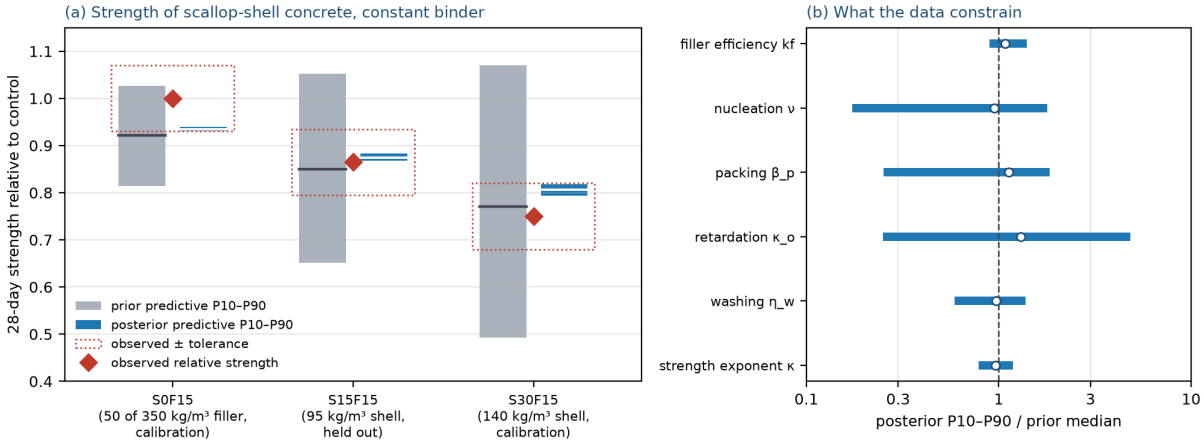


Fig. 3. Calibration against Abbas and Kudukkan (2025). (a) Relative 28-day strength: prior predictive with the k_f prior (grey), calibrated posterior predictive (blue) and observed values with the ±0.07 tolerance (red); S15F15 was not used in the calibration. (b) Posterior 10th–90th percentile range of selected parameters relative to their prior medians (circle: posterior median).

4.4.2 Independent oyster-shell data with frozen parameters

The Abbas-calibrated posterior was then frozen and used to predict three independent sets of oyster-shell data without refitting. (i) Kwon and Wang (2024): oyster-shell powder replacing 15% and 30% of cement in mortar at constant binder (568 kg/m³) and water-to-binder ratio 0.50 (Tables 1 and 3 of the source; observed relative strengths 0.65 and 0.43; particle size not reported, 30 µm assumed). (ii) Abinaya et al. as transcribed in Table A1 of Kim and Lee (2023): oyster shell at 2.5–10% replacement, water-to-binder ratio 0.40, observed relative strengths 1.05, 1.11, 1.05, 1.04; this is a secondary source, binder content (400 kg/m³) and particle size (30 µm) are assumed. (iii) De Dios-Suárez et al. (2026): untreated oyster-shell $CaCO_3$ at 10 wt% of cement in 1:4 mortar (water-to-cement ratio 0.68) for five sieve fractions of 0.14–0.84 mm (observed relative strengths 1.04, 0.98, 1.00, 1.01, 0.88); the 10 wt% is interpreted as replacement and the sieve apertures as particle sizes, both uncertain. Eleven data points result. The calibrated model is compared with the inert model, a k-value rule with $k_f = 0.25$ in the same model, and the Bolomey dilution relation $f \propto (c/w - 0.5)$ without and with a k-value of 0.25; the same errors are reported for all.

Table 6. Prediction of independent oyster-shell strength data (11 points, relative to each study's control) with frozen parameters: mean absolute error (MAE), root-mean-square error (RMSE), bias (prediction minus observation) and, for the draw-based models, the share of observations inside the 5–95% predictive interval widened by the 0.07 tolerance.

Model	MAE (all)	RMSE (all)	Bias (all)	90% interval coverage	MAE Kwon & Wang	MAE Abinaya	MAE mortar 2026
Abbas-calibrated, frozen	0.124	0.174	+0.019	64%	0.368	0.080	0.063
Inert filler ($k_f = 0$)	0.145	0.151	−0.094	27%	0.140	0.139	0.152
$k_f = 0.25$	0.133	0.146	−0.051	18%	0.225	0.116	0.110
Bolomey dilution	0.141	0.147	−0.083	–	0.159	0.141	0.134
Bolomey k-value 0.25	0.131	0.146	−0.045	–	0.234	0.121	0.098

The calibrated model has the lowest mean absolute error but not by a meaningful margin (0.124 against 0.145 for the inert model and 0.141 for dilution alone), and its root-mean-square error is higher than the inert model's (0.174 against 0.151) because of a few large misses. Its errors are not random (Fig. 4): it over-

predicts the large strength loss of Kwon and Wang by 0.37 (predicted 0.94 and 0.87 against 0.65 and 0.43) and is useful only where replacement is small or the filler is coarse, where its error for Abinaya et al. and the 2026 mortar study (0.080 and 0.063) is about half that of the inert model (0.139 and 0.152) and of dilution alone. The only open oyster-shell binary dataset with a large replacement loses more strength than pure dilution would, which no efficiency in the range 0–1 can reproduce. The calibrated model therefore does not outperform simple rules on independent data as a whole, and this is reported as a result: the calibration of one scallop-shell study is not evidence that the strength of oyster-shell concrete is predicted.

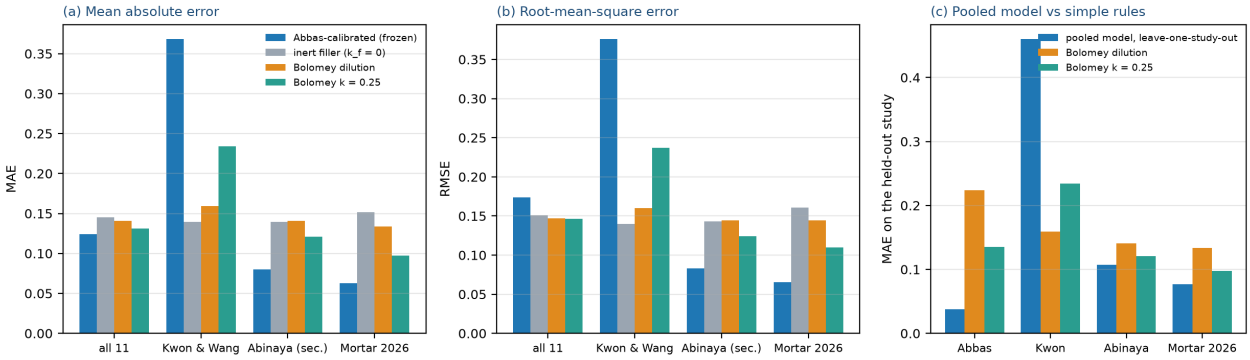


Fig. 4. Study-level prediction errors. (a) Mean absolute error and (b) root-mean-square error of the frozen Abbas-calibrated model, the inert model and two Bolomey rules on the 11 independent points, in total and by study. (c) Leave-one-study-out mean absolute error of the pooled model against the two Bolomey rules on each held-out study.

4.4.3 Study-level filler efficiency and the pooled predictive distribution

To describe what the four datasets say about a new study, k_f was estimated separately for each (rejection ABC on the study's mixes, k_f ~ U(−1.5, 1.5), tolerance 0.07; Table 7). The estimates differ in sign and size: -0.40 for Kwon and Wang (strength loss larger than dilution), 0.65 for Abbas and Kudukkan, 0.76 for the 2026 mortar study and 1.36 for Abinaya et al. (strength gains at low replacement). Treating the study estimates as a random sample, k_f for a new study has a Student-t predictive distribution (reference prior) with location 0.59, scale 0.82 and 3 degrees of freedom, truncated to [−1, 1.5] so that the effective cementing mass is not negative. This distribution, which expresses between-study heterogeneity and not a property of oyster shell, replaces the single posterior in all primary results below. In leave-one-study-out tests it predicts the held-out study with a mean absolute error of 0.038, 0.460, 0.108 and 0.077 for Abbas, Kwon, Abinaya and the 2026 mortar study, and its 90% interval covers every held-out point only because it is very wide.

Table 7. Study-level effective strength–efficiency coefficient k_f (rejection ABC, median and P10–P90) and leave-one-study-out prediction by the pooled distribution.

Study	k_f	Accepted draws	Leave-one-out MAE/bias	Evidence quality
Abbas and Kudukkan 2025 (scallop, concrete)	0.65 (0.54–0.85)	4414	0.038/−0.033	text values and derived moduli; three mixes
Kwon and Wang 2024 (oyster, mortar)	-0.40 (-0.65–-0.17)	23638	0.460/+0.460	tabulated values; two mixes; size unknown
Abinaya et al. via Kim and Lee 2023 (oyster)	1.36 (1.15–1.47)	17990	0.108/−0.108	secondary transcription; binder content assumed
De Dios-Suárez et al. 2026 (oyster $CaCO_3$, mortar)	0.76 (0.69–1.04)	31	0.077/−0.064	tabulated; coarse fractions; addition vs replacement unclear

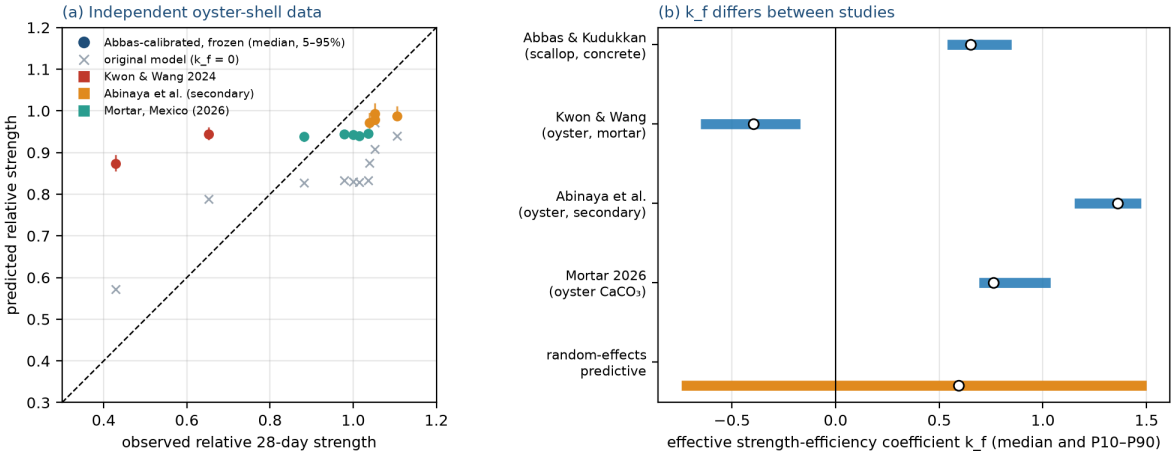


Fig. 5. Independent validation and study-level k_f. (a) Predicted against observed relative strength for 11 independent oyster-shell data points: Abbas-calibrated model with frozen parameters (coloured, median and 5–95% interval) and the original inert model (grey crosses). (b) k_f estimated separately for each study and the pooled random-effects predictive interval used for the primary results.

The calibration constrains strength only. Porosity, and hence chloride diffusivity, count cement only in the primary formulation and k_f is not applied to transport; no open dataset with tabulated chloride, acid or freeze–thaw results for shell-powder concrete was found, so the transport and damage predictions remain uncalibrated and are bracketed in Section 7.4. The study-level estimates condition on the other parameters being at their priors, ignore the weak posterior correlations among them, and the two weakest datasets (Abinaya, 2026 mortar) carry assumptions stated above; leaving them out leaves Abbas and Kwon with estimates of opposite sign and does not narrow the pooled interval.

4.5 Status of the model components

Strength is the only property for which any measured oyster-shell data enter the framework. To keep conditional results from being read as validated ones, each component is classified in Table 8.

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Table 8. Calibrated, uncalibrated and assumed components of the framework.

Component	Status	Basis and limit
Strength: k_f (effective strength efficiency)	Calibrated at study level; heterogeneous	Four studies, k_f from -0.40 to 1.36; pooled predictive distribution; one study is a secondary transcription
Strength: gel/space relation, hydration kinetics	Anchored to an OPC curve; generic ranges	Nominal OPC strength at w/c 0.45 about 43 MPa; shell-specific coefficients not measured
Chloride diffusivity of OPC	Anchored to the Life-365 relation	Fitted exponent reproduces the relation within a factor 1.23; no oyster-shell or limestone-filler measurements used
Chloride diffusivity of filler-bearing paste	Uncalibrated; bracketed	Two mappings (porosity and cement-only w/c) weighted by an uncertain parameter; porosity coupling of k_f tested (Section 7.4)
Acid and leaching fronts, freeze–thaw, fatigue, shrinkage, abrasion	Uncalibrated; phenomenological	Classical relations with generic parameter ranges; OPC reference lives set to plausible design lives, so results are relative
Carbon accounting	Generic factors; consistency check only	Budget reproduces the arithmetic of one paper (−24.4% and −36.0%) but is not independently validated; haul distance dominates carbon variance
Blended-cement reference	Screening treatment	Life-365 handling of slag; carbon and life not validated for this model
Service-life predictions	Hypothetical scenarios	Conditional on the uncalibrated components above; not a validated service-life model

5. Results

Sections 5–7 report the primary model, in which the strength-efficiency coefficient k_f follows the pooled random-effects predictive distribution of Section 4.4.3 (all other inputs unchanged). The original inert-filler model and the Abbas-only calibration are reported as ablations in Section 7.5.

5.1 Hydration and microstructure

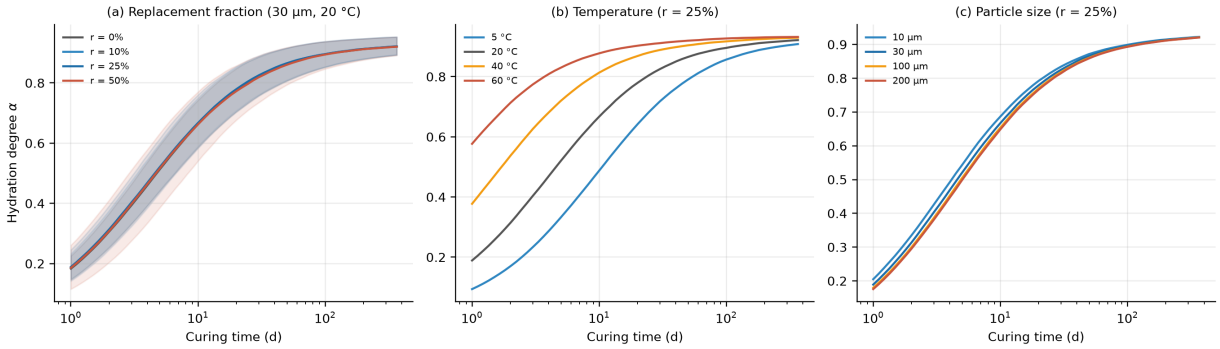


Fig. 6. Hydration degree against curing time (median and P10–P90): (a) shell replacement fraction; (b) curing temperature; (c) median particle size (P1, w/b = 0.42).

The model does not predict a large acceleration. Median hydration degree at 1 day is 0.18 for OPC and 0.19, 0.19 and 0.18 for 10%, 25% and 50% replacement; at 28 days the values are 0.81, 0.82, 0.82 and 0.81. The nucleation effect and the larger effective w/c are largely offset by organic retardation and dilution, so the probability that 1-day hydration is higher than OPC at r = 20% is 70% (the sign depends on the poorly known filler and retardation coefficients). Meanwhile cement is removed: median 28-day strength falls to 88% of OPC at r = 25% and 72% at r = 50%, and capillary porosity per unit paste rises by 46% and 94%.

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Temperature (Fig. 6b) accelerates hydration through the Arrhenius factor, and finer shell (Fig. 6c) accelerates it moderately.

5.2 Transport

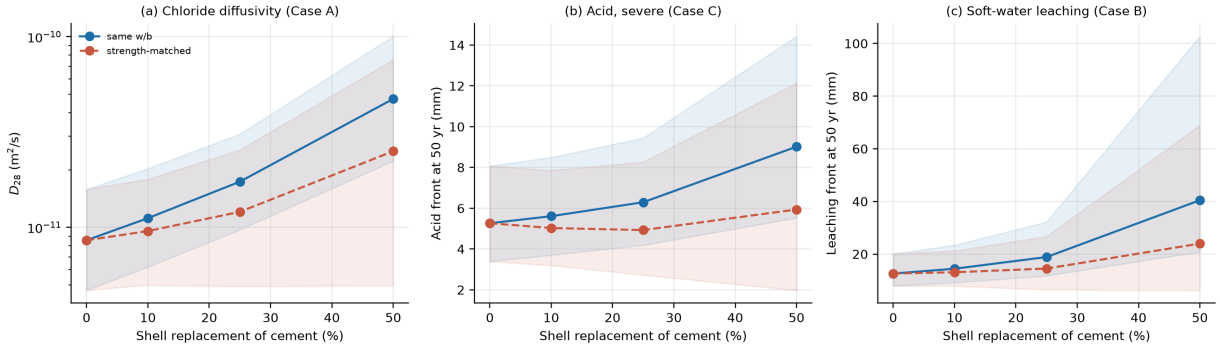


Fig. 7. Transport indicators against shell replacement for same-w/b and strength-matched mixes (median and P10–P90): (a) 28-day chloride diffusivity, pavement; (b) acid front depth at 50 years, severe exposure, culvert; (c) leaching front depth at 50 years, dam.

At constant w/b the median 28-day chloride diffusivity rises by a factor 1.3 at r = 10%, 2.1 at 25% and 5.6 at 50%, because dilution raises porosity and the cement-only mapping amplifies the effect. In the severe acid case the 50-year front is 5.3 mm for OPC and 6.3 mm at r = 25% (same w/b, +20%) but 4.9 mm when strength-matched (−7%): calcite buffering helps only when it is not outweighed by dilution. Particle size matters: at r = 25% the front is 6.2 mm for 10 µm shell and 7.3 mm for 200 µm shell, because coarse particles are less accessible. For soft-water leaching the 50-year front is 12.6 mm for OPC, 18.9 mm at r = 25% (+50%) and 40.6 mm at r = 50%, because calcite adds no leachable calcium and dilutes the cement. The calcite-dissolution mass loss in acid, which could offset the buffering benefit, is not modeled.

5.3 Service-life distributions

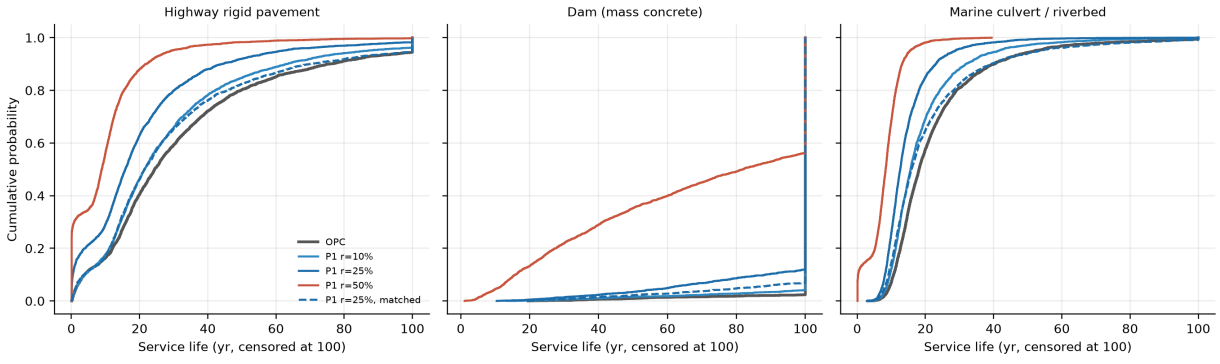


Fig. 8. Empirical cumulative distribution of service life (censored at 100 years) for OPC and selected shell mixes in the three cases (medium load, moderate exposure; 30 µm shell).

The OPC reference has a median life of 24.9 yr (P10–P90 4.5–70.4) in the pavement case, 18.1 yr (10.5–38.6) in the culvert case, and survives 100 years in 97% of dam draws. At constant w/b, shell replacement shifts the distributions to shorter life in the pavement and culvert cases, and strongly so at r ≥ 50% (Fig. 8). Strength-matched mixes recover part of the lost life (median −1.8 yr at r = 25%, pavement) but with a wide spread, because the filler efficiency differs between studies (Section 4.4.3) and the diffusivity mapping is untested (Section 7.3). Details by case are in Section 6.

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5.4 Carbon

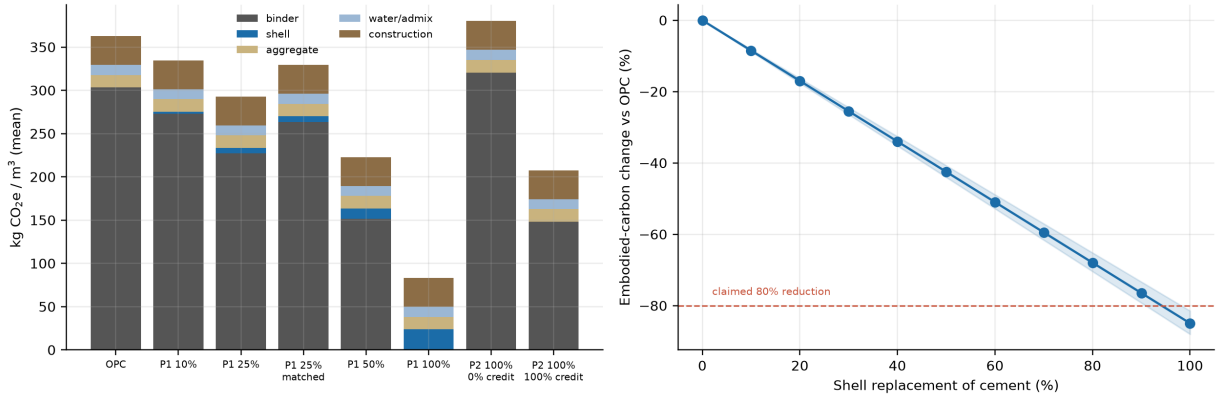


Fig. 9. Left: mean embodied carbon by component (pavement mix). Right: change in embodied carbon against shell replacement at constant w/b (median and P10–P90); the dashed line marks the 80% reduction claimed in the source white paper.

Table 9. Change in cradle-to-gate embodied carbon relative to OPC (median, P10–P90) and relative 28-day strength; pavement mix, 30 µm shell.

Configuration	Carbon change vs OPC (%)	28-day strength vs OPC
P1, r = 10%, same w/b	−8.5 (−8.8 to −8.1)	0.95
P1, r = 25%, same w/b	−21.2 (−22.0 to −20.4)	0.87
P1, r = 50%, same w/b	−42.5 (−44.0 to −40.7)	0.73
P1, r = 100%, same w/b	−84.9 (−88.1 to −81.5)	0.00
P1, r = 25%, strength-matched	−15.0 (−21.6 to +9.7)	1.02
P1, r = 50%, strength-matched	−37.0 (−43.3 to +13.0)	1.02
P2, ρ = 100%, no biogenic credit	+5.1 (+2.8 to +8.1)	n/a (≈ 1.0)
P2, ρ = 100%, 50% biogenic credit	−21.0 (−24.0 to −17.7)	n/a (≈ 1.0)
P2, ρ = 100%, 100% biogenic credit	−47.1 (−51.7 to −42.7)	n/a (≈ 1.0)

At constant w/b, carbon falls almost proportionally to r (−8.5%, −21.2%, −42.5% at 10, 25 and 50%) because the shell powder footprint is small relative to cement. The claimed 80% reduction is reached only at essentially complete cement replacement (−85.0% at r = 100%), at which the concrete has no strength. When strength is matched by adding binder, the median benefit shrinks to −15.1% at r = 25% and −36.7% at r = 50%, with a wide interval (P10–P90 −21.6 to +9.7% at 25%) that includes net increases where the effective efficiency is low or negative. Pathway P2 increases embodied carbon by 5.1% (P10–P90 2.8 to 8.1) when no biogenic credit is allowed, because processing and transport of shell exceed the footprint of quarrying adjacent limestone while the calcination CO_2 is unchanged; a reduction appears only when 50% or 100% of the calcination CO_2 is credited (-21.0% and -47.1%), which is an accounting choice and not a material property.

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5.5 Carbon–durability trade-off

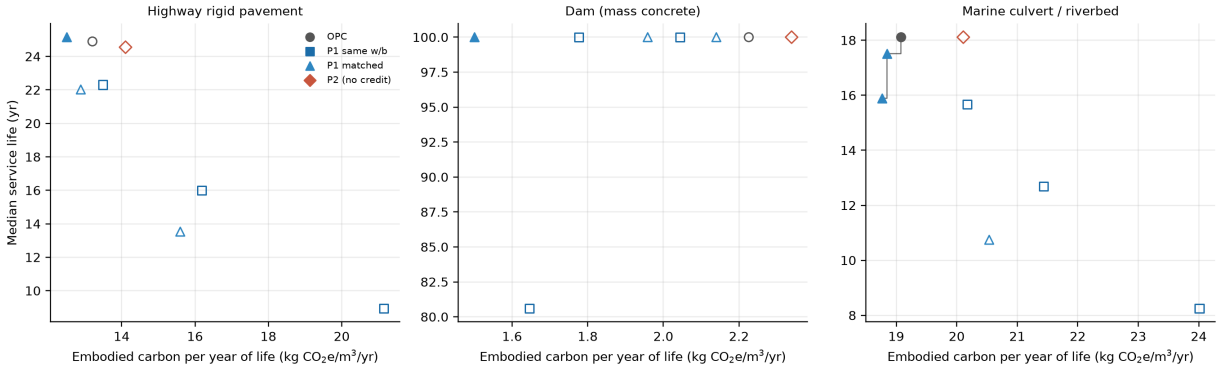


Fig. 10. Embodied carbon per year of life against median service life for OPC, P1 ($r \leq 50\%$, 30 µm; same w/b and strength-matched) and P2 without credit. Filled symbols are Pareto-optimal, open symbols are dominated.

Over the plausible replacement range ($r \leq 50\%$), no shell mix dominates OPC in every case. Strength-matched P1 mixes cut carbon by a median −15% at r = 25% but lose a median 1.8 yr (pavement) and 2.3 yr (culvert) of life, whereas same-w/b mixes cut carbon by −21% and lose 8.4 and 5.5 yr; in the dam case life is bounded by the 100-year horizon (Fig. 10). These are median-based statements; the joint-advantage probabilities in Section 7 give the more informative picture.

6. Infrastructure applications

6.1 Case A: Highway rigid pavement

The pavement case combines repeated traffic loading, restrained shrinkage and curling, freeze–thaw and chloride ingress from de-icing salt (medium stress ratio, moderate exposure). The OPC failure mode is chloride corrosion initiation plus propagation in 73% of draws and damage in 23% (Table 10).

Table 10. Case A results (30 µm shell; 3,000 paired draws; medium stress ratio, moderate chloride exposure).

Mix	f_c28 (MPa)	Life P10/P50/P90 (yr)	ΔL P50 (yr)	P(noninferior)	ΔE P50 (%)	P(≥5% cut & noninf.)
OPC reference	49	4.5/24.9/70.4	–	–	–	–
P1 10%, same w/b	46	5.2/22.3/58.5	−2.7	18%	−8.5	18%
P1 25%, same w/b	42	0.5/16.0/41.6	−8.4	14%	−21.2	14%
P1 50%, same w/b	35	0.1/8.9/20.7	−16.6	11%	−42.5	11%
P1 10%, matched	49	6.4/25.2/73.5	−0.1	48%	−6.2	15%
P1 25%, matched	49	5.1/22.0/66.7	−1.8	41%	−15.0	21%
P1 50%, matched	49	0.1/13.5/53.8	−8.2	30%	−37.0	19%
P2 100%, no credit	47	4.0/24.6/69.6	0.0	4%	+5.1	0%
P2 100%, 50% credit	47	4.0/24.6/69.6	0.0	4%	−21.0	4%

At constant w/b, replacement reduces median life by 2.7 yr (10%), 8.4 yr (25%) and 16.6 yr (50%), since the weaker, more porous paste fatigues and admits chloride faster. The restrained-shrinkage cracking index exceeds 1 in 44% of OPC draws and in 37%, 24% and 5% of draws at 10%, 25% and 50% replacement (same w/b), a shrinkage benefit that arises from dilution; with strength matched it is 35%. Strength-matched mixes change median life by −1.8 yr at r = 25% (P10–P90 −20.6 to +15.8) and carbon by −15.0%.

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Sensitivity of the life difference is led by strength-efficiency coefficient k_f (0.42), restraint factor (0.23), relative humidity (0.20).

6.2 Case B: dam (mass concrete)

For mass concrete the modeled mechanisms (thermal cracking, leaching with hydrostatic gradient, freeze–thaw on the face) rarely end the life of the OPC reference within 100 years (97% survive), so differences in life are censored and the damage-index tie-break governs (Table 11). Lower cement content reduces the adiabatic temperature rise, but the OPC thermal cracking probability is already 0%, so little is gained. At r = 10% (same w/b) the probability of non-inferior life is 55% and that of a joint 5% carbon cut and non-inferior life is 55%; at 25% it falls to 5% because the leaching front deepens (Section 5.2), and at 50% it is 0%. The apparent success at low replacement is therefore a statement that dam concrete is durable in the model regardless, not evidence of benefit; with strength matched at r = 25% it is 14% (−14.2% carbon). The reference here is OPC although dam concrete usually contains supplementary cementitious materials. Life differences are driven by air-void filling fraction (0.35), permeability factor (0.33), critical degree of saturation (0.30).

Table 11. Case B results (30 µm shell; 3,000 paired draws; moderate exposure). Median life is censored at 100 years for the OPC reference.

Mix	f_c28 (MPa)	Life P10/P50/P90 (yr)	ΔL P50 (yr)	P(noninferior)	ΔE P50 (%)	P(≥5% cut & noninf.)
OPC reference	34	100.0/100.0/100.0	–	–	–	–
P1 10%, same w/b	32	100.0/100.0/100.0	0.0	55%	−8.1	55%
P1 25%, same w/b	29	94.2/100.0/100.0	0.0	5%	−20.2	5%
P1 50%, same w/b	24	16.2/80.6/100.0	−18.9	0%	−40.5	0%
P1 10%, matched	36	100.0/100.0/100.0	0.0	77%	−5.7	33%
P1 25%, matched	36	100.0/100.0/100.0	0.0	46%	−14.2	14%
P1 50%, matched	36	28.6/100.0/100.0	0.0	29%	−34.6	6%
P2 100%, no credit	34	100.0/100.0/100.0	0.0	100%	+4.9	0%
P2 100%, 50% credit	34	100.0/100.0/100.0	0.0	100%	−20.0	100%

6.3 Case C: marine culvert / riverbed

The culvert case combines chloride ingress, acid from organic matter and abrasion. OPC failure is by chloride corrosion in 96% of draws (Table 12). At constant w/b the median life falls by 2.5 yr at 10%, 5.5 yr at 25% and 10.8 yr at 50%, and the probability of non-inferior life is 0%, 0% and 0%: shell concrete underperforms OPC in essentially every draw when strength is not matched. Strength matching reduces the median loss (−2.3 yr at 25%) and gives non-inferior life in 33% of draws, but the joint-advantage probability is at most 3% because the draws with non-inferior life are mostly those in which the matched mix needs so much added cement that the carbon cut is below 5%. The acid benefit of calcite buffering (Section 5.2) is too small relative to chloride-driven life to alter the failure mode. Life differences are driven by chloride-diffusivity multiplier (0.35), surface chloride (0.17), service temperature (0.17).

Table 12. Case C results (30 µm shell; 3,000 paired draws; moderate exposure).

Mix	f_c28 (MPa)	Life P10/P50/P90 (yr)	ΔL P50 (yr)	P(noninferior)	ΔE P50 (%)	P(≥5% cut & noninf.)
OPC reference	49	10.5/18.1/38.6	–	–	–	–
P1 10%, same w/b	46	9.5/15.7/31.0	−2.5	0%	−8.5	0%
P1 25%, same w/b	42	8.1/12.7/22.0	−5.5	0%	−21.3	0%
P1 50%, same w/b	35	0.3/8.3/12.8	−10.8	0%	−42.7	0%
P1 10%, matched	49	10.1/17.5/38.0	−1.0	33%	−6.2	0%
P1 25%, matched	49	9.0/15.9/40.2	−2.3	33%	−15.1	3%
P1 50%, matched	49	3.3/10.7/34.5	−6.2	21%	−37.2	4%
P2 100%, no credit	47	10.5/18.1/38.6	0.0	96%	+5.1	0%
P2 100%, 50% credit	47	10.5/18.1/38.6	0.0	96%	−21.1	96%

6.4 Comparison with a low-clinker blended-cement reference

Plain OPC is not what engineers would choose today for marine or mass concrete. A blended-cement reference was therefore added: 50% of the binder replaced by ground granulated blast-furnace slag, with the Life-365 treatment of slag (age exponent of the diffusivity raised by 0.4 × 0.5/0.7, 28-day diffusivity at equal water-to-binder ratio unchanged), a slag emission factor of 0.07 kg CO_2e/kg (0.12 as a sensitivity), identical 28-day strength and the same acid and leaching treatment as OPC; this is a screening treatment and not a validated slag-concrete model. The blended reference cuts embodied carbon by 42% (P10–P90 41–43%) and raises the median life from 24.4 to 96.2 yr in the pavement case and from 18.2 to 68.2 yr in the culvert case (Table 13).

Table 13. Shell concrete (P1, r = 25%, 30 µm, primary model) against plain OPC and against the 50% slag blended-cement reference: median change in embodied carbon and in service life (P10–P90) and P (≥5% carbon cut and noninferior life).

Comparison	ΔE median (P10 to P90) %	ΔL median (P10 to P90) yr	P(adv, mat)
Pavement: blended reference vs OPC	−42	+34.2	–
Pavement: shell 25%, same w/b vs OPC	−21 (−22 to −20)	−8.0 (−32.2 to +2.0)	0.15
Pavement: shell 25%, matched vs OPC	−15 (−22 to +10)	−1.8 (−19.6 to +12.9)	0.21
Pavement: shell 25%, matched vs blended	+47 (+35 to +89)	−40.2 (−79.9 to +4.7)	0.00
Pavement: shell 50%, matched vs blended	+10 (−2 to +95)	−49.2 (−91.0 to +6.2)	0.00
Dam: blended reference vs OPC	−40	0.0	–
Dam: shell 25%, same w/b vs OPC	−20 (−21 to −19)	0.0 (−4.6 to 0.0)	0.04
Dam: shell 25%, matched vs OPC	−14 (−21 to +23)	0.0 (0.0 to 0.0)	0.13
Dam: shell 25%, matched vs blended	+44 (+32 to +106)	0.0 (0.0 to 0.0)	0.00
Dam: shell 50%, matched vs blended	+9 (−2 to +111)	0.0 (−70.8 to 0.0)	0.00
Culvert: blended reference vs OPC	−42	+43.1	–
Culvert: shell 25%, same w/b vs OPC	−21 (−22 to −20)	−5.6 (−17.2 to −2.0)	0.00
Culvert: shell 25%, matched vs OPC	−15 (−22 to +10)	−2.2 (−12.1 to +9.9)	0.03
Culvert: shell 25%, matched vs blended	+48 (+36 to +90)	−42.3 (−82.3 to −6.7)	0.00
Culvert: shell 50%, matched vs blended	+10 (−2 to +95)	−48.2 (−90.2 to −6.8)	0.00

Against the blended reference, shell concrete at r = 25% has carbon higher by a median 47% in the pavement case (strength matched) and a shorter life, and the joint-advantage probability is 0.00 in all three cases at r = 25% and 50%. A realistic low-clinker concrete already delivers the carbon cut that shell replacement offers, with better chloride durability; in the framework, shell powder at the replacement levels studied here is not a competitor to slag blends but at best a complement, and its value must be sought where slag is not available or in the shell's other uses (Sandler 2026b).

Table 14. Sensitivity to exposure severity and load class for P1 at r = 25% (30 µm): median life and P (≥5% carbon cut and noninferior life), same w/b and strength-matched.

Case and condition	OPC life P50	Same w/b: life P50	P(adv, mat)	Matched: life P50	P(adv, mat)
A: moderate exposure, low load	29.1	19.2	0.06	26.5	0.11
A: moderate exposure, medium load	24.6	15.6	0.15	21.4	0.21
A: moderate exposure, high load	14.2	8.4	0.29	14.0	0.39
A: mild exposure, medium load	48.9	28.3	0.30	44.7	0.36
A: severe exposure, medium load	17.0	11.7	0.12	15.1	0.17
B: mild exposure, medium load	100.0	100.0	0.14	100.0	0.21
B: moderate exposure, medium load	100.0	100.0	0.04	100.0	0.14
B: severe exposure, medium load	100.0	100.0	0.02	100.0	0.10
C: mild exposure, medium load	31.5	18.9	0.01	26.5	0.05
C: moderate exposure, medium load	18.3	12.6	0.00	15.8	0.02
C: severe exposure, medium load	13.8	10.4	0.00	12.4	0.02

Across exposure severity and load class the joint-advantage probability for P1 at r = 25% ranges from 0.00 to 0.30 for same-w/b mixes and from 0.02 to 0.39 for strength-matched mixes; no condition gives a high probability, and the highest values occur where the OPC reference is censored at the horizon (dam) or the exposure is mild.

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7. Uncertainty and falsification

7.1 Falsification maps

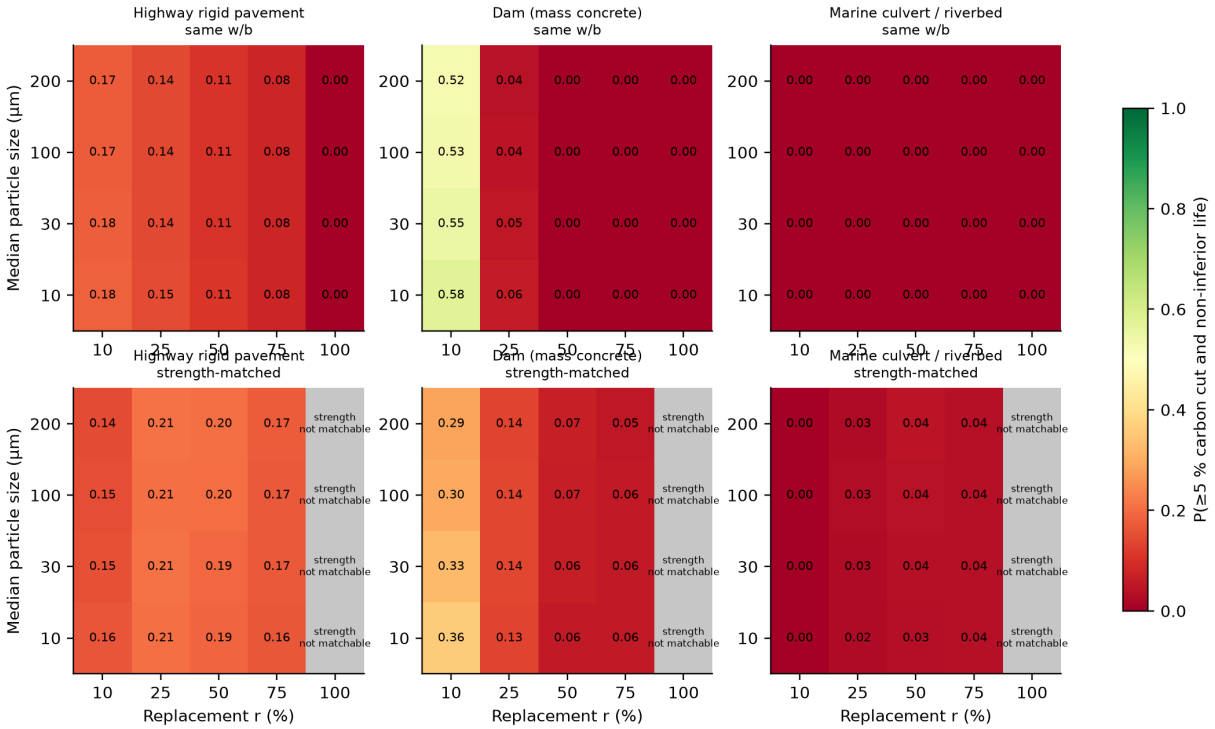


Fig. 11. Probability of a joint advantage (carbon cut of at least 5% and non-inferior service life) for P1 as a function of replacement fraction and median particle size, same-w/b (top) and strength-matched (bottom) designs. Grey cells mark strength-matched designs for which matching is infeasible in more than half of the draws.

The joint-advantage region is small. For the pavement and culvert cases the same-w/b maps reach at most 0.18 and 0.00 for r ≤ 50%, and the strength-matched maps reach up to 0.21 for the pavement and 0.04 for the culvert, wherever matching is possible in at least half of the draws. The dam map is non-empty at r = 10% mainly for the censoring reason explained in Section 6.2. With the strict definition (any carbon reduction), matched mixes appear to succeed in 26% of pavement draws at r = 25%, but the carbon change is −15.0% (P10–P90 −21.6 to +9.7%), so the material threshold of 5% is used.

7.2 Pathway P2

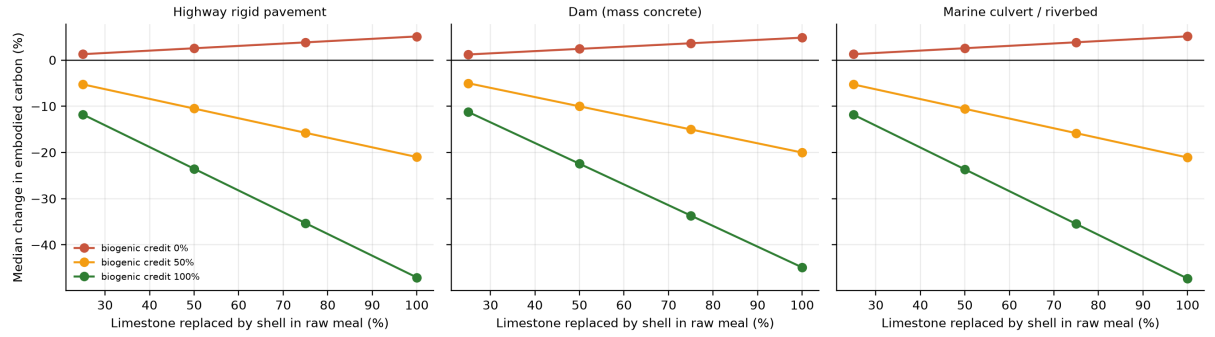


Fig. 12. P2 pathway: median change in embodied carbon against the fraction of limestone replaced by shell in raw meal for three levels of biogenic credit.

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For P2 the durability effect is negligible (median life change −0.04 yr in the pavement case from the impurity penalty, none in the dam and culvert cases), so the outcome is determined by carbon accounting. Without credit P2 is falsified: embodied carbon is higher in 100% of draws. With a 50% or 100% biogenic credit, carbon falls by 21.0% and 47.1% and the joint advantage is reached in 96%–100% of dam and culvert draws (for the pavement, the strict non-inferiority test fails in most draws because of the small strength penalty; with a 5% life tolerance the probability is 81%). Whether shell calcination CO_2 may be treated as biogenic is a methodological and regulatory question that this model cannot resolve; the credit should be reported as a scenario, not a property.

7.3 Model-form uncertainty and sensitivity

Table 15. Dependence of the median life difference (ΔL, years) for P1 at r = 25% on the diffusivity model weight χ_D (0 = porosity mapping, 1 = cement-only w/c mapping).

χ_D range	Pavement: same w/b	Pavement: matched	Culvert: same w/b	Culvert: matched
0.00–0.25	−6.7	−0.6	−5.3	−1.5
0.25–0.50	−8.2	−1.4	−5.5	−2.1
0.50–0.75	−7.8	−2.1	−5.7	−2.5
0.75–1.00	−9.1	−3.0	−5.6	−2.6

The sign of the same-w/b effect is robust to the diffusivity mapping (negative in all bins), whereas the median life difference of the matched design ranges from −0.6 yr to −3.0 yr in the pavement case as the pessimistic mapping is weighted more heavily. The matched-design result also depends on the assumed filler shift ω of the ageing exponent, which favors filler-bearing mixes and has no experimental support here.

Because no transport data for shell-filler concrete are available, service life was also evaluated under fixed transport models instead of the sampled weight: the porosity mapping alone (χ_D = 0, filler refines the pores), the cement-only mapping alone (χ_D = 1, filler inert for transport), each with and without the porosity coupling of k_f (Table 16). The joint-advantage probability for the pavement with strength matching ranges from 0.18 to 0.68, and that for the marine culvert from 0.01 to 0.66. The durability conclusion thus spans an interval set by assumptions that are not measured, and the favorable end requires both pore refinement by the shell and a diffusivity that follows porosity, which is the least supported combination.

Table 16. Joint advantage probability P(adv, mat) for P1 at r = 25% (30 µm, medium load, moderate exposure, 3,000 paired draws) under fixed transport models; same w/b/strength matched.

Transport model	Pavement	Dam	Culvert
χ_D = 0.0, strength-only k_f	0.17/0.26	0.17/0.25	0.00/0.05
χ_D = 0.0, porosity-coupled k_f	0.54/0.68	0.54/0.63	0.43/0.66
χ_D = 0.5, strength-only k_f	0.14/0.21	0.02/0.12	0.00/0.02
χ_D = 0.5, porosity-coupled k_f	0.33/0.48	0.13/0.28	0.14/0.23
χ_D = 1.0, strength-only k_f	0.14/0.18	0.00/0.06	0.00/0.01
χ_D = 1.0, porosity-coupled k_f	0.14/0.18	0.01/0.07	0.01/0.02

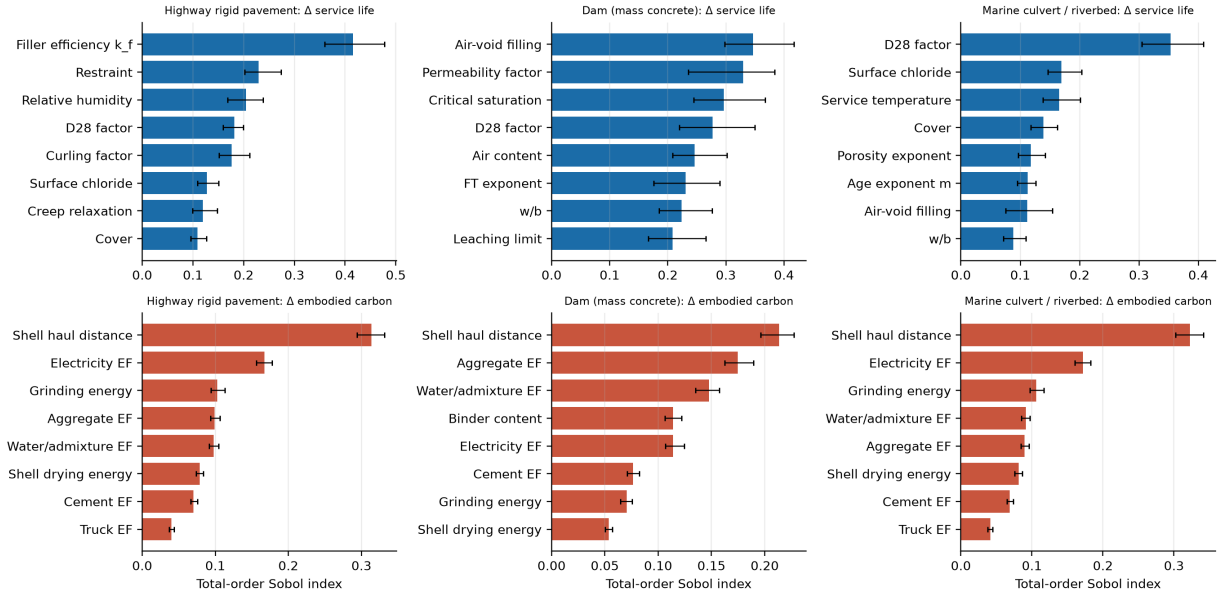


Fig. 13. Total-order Sobol indices (eight largest, with 95% bootstrap intervals) for the difference in service life (top) and in embodied carbon (bottom) between P1 at r = 25% and OPC, in each case.

The life difference is controlled by strength-efficiency coefficient k_f (0.42), restraint factor (0.23), relative humidity (0.20), chloride-diffusivity multiplier (0.18) in the pavement case, by air-void filling fraction (0.35), permeability factor (0.33), critical degree of saturation (0.30), chloride-diffusivity multiplier (0.28) in the dam case, and by chloride-diffusivity multiplier (0.35), surface chloride (0.17), service temperature (0.17), cover depth (0.14) in the culvert case; the sums of the total-order indices exceed one, indicating interactions. The first-order Jansen estimates were noisy (the sums of first-order indices for the three life-difference analyses were 3.19, 0.98 and 0.05, whereas a stable estimate lies between 0 and 1) because of the many parameters and the censored, tie-heavy output, and are not reported. The carbon difference is controlled by logistics and energy inputs: shell haul distance (0.31), electricity emission factor (0.17), grinding energy (0.10). The shell haul distance alone explains 31% of the carbon variance for the pavement, which means that carbon claims for shell concrete cannot be separated from the supply-chain geography.

7.4 Alternative filler–porosity–transport formulations

The primary formulation applies k_f to strength only. Two alternatives were tested with the same pooled distribution: k_f also refines the capillary porosity (and therefore the chloride diffusivity and the freeze–thaw saturation), and k_f scales with particle fineness. The size-dependent formulation fits worse: with it, no prior draw reproduces the Abbas and Kudukkan calibration mixes within the tolerance (best-fit k_f 0.70, near the edge of the prior) or the 2026 mortar data (best-fit 1.50, at the bound), against 4,414 accepted draws for the strength-only form, and it is not carried further. The porosity-coupled form leaves strength unchanged but changes the durability conclusions (Table 17): the pavement joint-advantage probability for the matched design at r = 25% rises from 0.21 to 0.45, the dam value from 0.14 to 0.32 and the culvert value from 0.03 to 0.29. The durability conclusion is therefore as sensitive to the unmeasured coupling between the filler and the pore structure as it is to the strength calibration, and the primary results are the conservative end of a bracket.

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Table 17. Strength-only (primary) and porosity-coupled formulations of k_f: P (≥5% carbon cut and noninferior life), median life difference (yr) and median 28-day chloride diffusivity ratio to OPC (r = 25% same w/b).

Case and design	P(adv, mat): strength-only/coupled	ΔL P50 (yr)	D_{28} ratio P50
Pavement: 25% same w/b	0.14/0.33	−8.4/−3.7	2.05/1.80
Pavement: 25% matched	0.21/0.45	−1.8/+1.8	–
Pavement: 50% matched	0.19/0.40	−8.2/0.0	–
Dam: 25% same w/b	0.05/0.20	0.0/0.0	1.97/1.84
Dam: 25% matched	0.14/0.32	0.0/0.0	–
Dam: 50% matched	0.06/0.17	0.0/0.0	–
Culvert: 25% same w/b	0.00/0.18	−5.5/−3.6	2.05/1.80
Culvert: 25% matched	0.03/0.29	−2.3/+0.2	–
Culvert: 50% matched	0.04/0.26	−6.2/−1.8	–

Because the pooled distribution rests on four studies, the whole analysis was repeated four times with one study left out of the pooling, so that the service-life and carbon conclusions, not only the strength predictions, can be checked for dependence on one dataset (Table 18). The pavement joint-advantage probability at r = 25% with strength matching stays between 0.18 and 0.24 (all four studies: 0.21), the dam probability between 0.11 and 0.15 and the culvert probability between 0.02 and 0.03; the median carbon cut moves from −9% to −20%. The qualitative conclusion, a modest joint-advantage probability for the pavement and the dam and a negligible one for the culvert, does not depend on any one study, although the carbon cut does: leaving out Abinaya et al. (the study with the highest efficiency) lowers it and leaving out Kwon and Wang (the lowest) raises it.

Table 18. Leave-one-study-out analysis of the P1 results at r = 25% with strength matching (30 µm): pooled k_f distribution of the remaining studies, median carbon change (%), median life difference (yr) and P(adv, mat) for the pavement, dam and culvert.

Study left out	k_f location/scale	Pavement: ΔE, ΔL, P	Dam: ΔE, ΔL, P	Culvert: ΔE, ΔL, P
None (all four studies, primary)	0.59/0.82	−15%, −1.8, 0.21	−14%, 0.0, 0.14	−15%, −2.3, 0.03
Abbas and Kudukkan	0.57/1.03	−15%, −2.5, 0.19	−14%, 0.0, 0.11	−15%, −2.0, 0.02
Kwon and Wang	0.92/0.44	−20%, −3.5, 0.24	−19%, 0.0, 0.13	−20%, −3.9, 0.02
Abinaya et al.	0.34/0.74	−9%, −1.0, 0.19	−9%, 0.0, 0.15	−9%, −1.1, 0.03
Mortar 2026	0.54/1.02	−14%, −2.4, 0.18	−13%, 0.0, 0.11	−14%, −1.9, 0.02

7.5 Ablation: original model, single-study calibration and pooled calibration

The comparison with earlier versions of the model is given in Table 19 and Fig. 14. With shell as an inert filler (the original model) the pavement joint-advantage probability at r = 25% is 0.01 (same w/b) and 0.01 (matched), and strength-matched carbon cuts are −0.3%; calibrated on Abbas and Kudukkan alone they are 0.23, 0.28 and −16.4%; with the pooled distribution 0.14, 0.21 and −15.0%. The conclusion that a joint advantage is rare holds in all three, but its size depends on the filler efficiency, which is the least certain strength parameter and is not a property of the material alone.

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Table 19. Ablation of the P1 results across model variants (r = 25%, 30 µm): median carbon change (%), median 28-day strength relative to OPC and P (≥5% carbon cut and noninferior life): original (k_f = 0)/Abbas-calibrated/pooled (primary).

Case and design	ΔE P50 (%)	f_c28 vs OPC	P(adv, mat)
Pavement: 25% same w/b	−21/−21/−21	0.67/0.91/0.87	0.01/0.23/0.14
Pavement: 25% matched	0/−16/−15	1.00/1.00/1.02	0.01/0.28/0.21
Dam: 25% same w/b	−20/−20/−20	0.64/0.90/0.85	0.04/0.04/0.05
Dam: 25% matched	0/−15/−14	1.00/1.00/1.04	0.00/0.19/0.14
Culvert: 25% same w/b	−21/−21/−21	0.67/0.91/0.87	0.00/0.00/0.00
Culvert: 25% matched	0/−16/−15	1.00/1.00/1.02	0.01/0.00/0.03

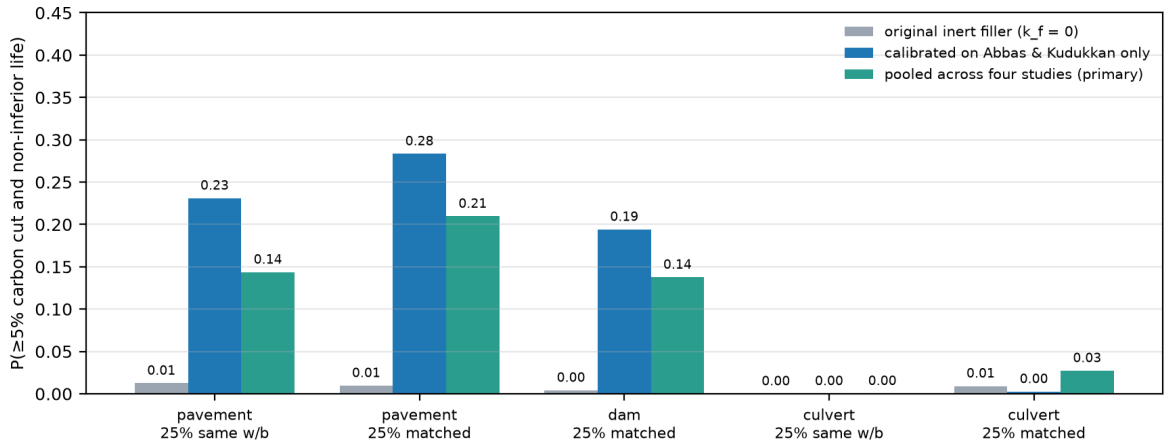


Fig. 14. Probability of a joint advantage (carbon cut of at least 5% and noninferior life) for P1 at r = 25% (30 µm) under three model variants: original inert filler, calibration on a single scallop-shell study, and the pooled random-effects distribution used for the primary results.

7.6 Summary of the joint advantage probabilities and their conditions

The probabilities quoted in the abstract, results and conclusions refer to different model configurations; Table 20 lists each with its conditions. All values are for 25% replacement of cement by 30 µm shell, medium load, moderate exposure and 3,000 paired draws, with the joint advantage defined as a carbon cut of at least 5% and non-inferior life; the first column of each pair uses the same water-to-binder ratio and the second the strength-matched design.

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Table 20. Joint advantage probability P(adv, mat) for P1 at r = 25% under each model configuration: same w/b/strength matched, by case. The reference binder is OPC except in the last row.

Configuration (k_f model; transport assumption; reference)	Pavement	Dam	Culvert
Primary: pooled k_f, strength-only; sampled diffusivity mapping; OPC	0.14/0.21	0.05/0.14	0.00/0.03
Porosity-coupled k_f; sampled mapping; OPC	0.33/0.45	0.20/0.32	0.18/0.29
Strength-only; porosity mapping only (χ_D = 0); OPC	0.17/0.26	0.17/0.25	0.00/0.05
Strength-only; cement-only mapping (χ_D = 1); OPC	0.14/0.18	0.00/0.06	0.00/0.01
Porosity-coupled; porosity mapping only (χ_D = 0); OPC	0.54/0.68	0.54/0.63	0.43/0.66
Original inert filler (k_f = 0); sampled mapping; OPC	0.01/0.01	0.04/0.00	0.00/0.01
Calibrated on Abbas and Kudukkan only; sampled mapping; OPC	0.23/0.28	0.04/0.19	0.00/0.00
Primary model against 50% slag blended cement	0.00/0.00	0.00/0.00	0.00/0.00

The values quoted as maxima in the abstract and conclusions (0.21 for the pavement) are the first row; the value of 0.45 quoted for the pore-structure assumption is the second row, from the same draws. The two are not contradictory: they are the same design under different transport assumptions. None is an empirical probability for real infrastructure: each is the share of Monte Carlo draws drawn from author-assigned priors and untested transport assumptions, in which the stated criterion is met, and the shell-specific chloride diffusivity, acid attack, freeze–thaw and service-life relationships are uncalibrated (Table 8).

8. Discussion

8.1 Physical interpretation

The main results follow from three mechanisms. First, shell powder is a diluent: removing cement lowers strength and raises porosity almost in proportion, so any benefit must come from a filler effect. The size of that effect is the least certain strength parameter, and the calibration does not generalize across shell materials (Section 4.4): it is -0.40 to 1.36 across four studies, and it moves the pavement joint-advantage probability from about 0.01 (inert filler) to 0.14–0.21 (pooled) at r = 25% (Section 7.5). Second, embodied carbon is dominated by cement, so replacing cement lowers carbon in proportion to r at constant w/b; at constant strength the added binder re-introduces cement, and the median carbon cut falls from −21% to −15% at r = 25%, with an interval that includes net increases. Third, the carbon benefit of P2 is tied to a convention (biogenic credit), since the calcination CO_2 per kg of $CaCO_3$ is fixed by stoichiometry. The dam and culvert results show where shell concrete might be advantageous (low replacement in mass concrete) and where it is not (chloride-controlled marine exposure), and the blended-cement comparison (Section 6.4) shows that a conventional low-clinker concrete already delivers a larger carbon cut than shell replacement at the levels modeled.

8.2 Assessment of the white paper's claims

Table 21. Source claims restated as hypotheses with model outputs and the validation each would need.

Claim	Model output (this study)	Status	Validation needed
Up to 80% CO_2 reduction	−85.0% only at r = 100% (no strength); −21.2%/−42.5% at r = 25/50% (same w/b); −15.1%/−36.7% matched; P2 +5.1% without credit	Not supported	Plant-specific cradle-to-gate LCA with defined boundary and functional unit
Thermal resistance to 2,400 °F	Not modeled	Not supported (see text)	Furnace tests with residual strength; thermogravimetry of the shell binder
100% limestone substitution	P2, ρ = 100%: carbon +5.1% (no credit); durability unchanged	Feasible chemically; no carbon benefit without credit	Raw-meal burnability, clinker chemistry, chloride and alkali limits
Accelerated hydration	α(1 d): 0.18 (OPC) vs 0.19 (r = 25%); strength 88% of OPC	Marginal acceleration; strength lower	Isothermal calorimetry; setting time; strength at 1–365 d
Reduced shrinkage	Cracking probability 44% → 24% at r = 25% (same w/b); 35% when matched	Supported only as dilution	Restrained-ring and free-shrinkage tests
Superior acid resistance	50-yr front +20% (same w/b) vs −7% (matched); particle-size dependent	Conditional	Acid immersion with mass loss and depth of degradation
Higher structural durability	ΔL P50 −8.4 yr (pavement) and −5.5 yr (culvert) at r = 25% same w/b; −1.8/−2.3 yr matched	Not supported at same w/b; conditional if matched	Chloride migration, freeze–thaw, fatigue and abrasion tests on matched mixes

On the thermal claim. 2,400 °F is about 1,316 °C. Calcite decomposes to CaO and CO_2 at roughly 900 °C (about 1,650 °F) at atmospheric CO_2 pressure (standard thermodynamic data; Chase 1998), so a calcium-carbonate-rich binder is not expected to withstand 2,400 °F; even conventional concrete suffers substantial strength loss and spalling at much lower temperatures. No model of fire behavior is attempted here and the claim should be removed unless supported by formulation-specific tests.

8.3 Limitations

- No new experiments are reported. Shell-specific parameters (ν, β_p, κ_o, ω, k_imp) are literature-style ranges; only the strength model was calibrated, on one published scallop-shell dataset (three mixes, replacement 14–40% of the binder) whose strengths are partly read from text and derived from moduli, with the held-out mix between the calibration mixes. Independent testing used 11 oyster-shell points from three studies, one of them a secondary transcription and one with coarse particle fractions and an unclear replacement definition; k_f differs in sign and size between studies, so the pooled distribution describes heterogeneity and not a material property, and it ignores correlations with the other parameters.
- Chloride, acid and freeze–thaw transport remain uncalibrated: paste porosity counts cement only, the filler efficiency is not applied to transport, and the claims that depend on life (Section 7.4) are therefore bracketed rather than determined.
- The OPC reference lives are anchored to plausible design lives by adjusting OPC-only parameters; results are relative, and the absolute lives are not predictions.
- The diffusivity mapping for filler-bearing pastes is an extrapolation of a relation calibrated for OPC; the two mappings bracket but do not validate it.
- The reference is plain OPC concrete; dam and marine concretes normally contain supplementary cementitious materials, which would change both carbon and durability baselines.

- Mechanisms are combined additively and one-dimensionally; carbonation, alkali–silica reaction, sulfate attack, reinforcement corrosion kinetics, calcite dissolution mass loss and the interaction of cracking with all transport processes are not modeled; surface chloride is constant.
- Carbon factors are generic screening values; the avoided-landfill credit, shell logistics and biogenic accounting are uncertain and dominate the carbon variance.
- The strength-matching rule (adding binder at constant water) is only one of several ways to achieve equal strength and is not applicable where the model strength saturates with binder.

8.4 Novelty and validation requirements

The contribution is not the finding that fine calcium carbonate acts as a filler, which is established for limestone, but a framework that couples processing-dependent carbon with hydration, transport and damage, evaluates each shell mix against its own OPC reference under shared uncertainty, and reports where the joint advantage fails. To be a validated model, the following would be needed: (i) isothermal calorimetry, porosimetry and strength development on shell-powder pastes and concretes at several replacement levels, particle sizes and organic contents; (ii) chloride migration/diffusion, acid-immersion, freeze–thaw, fatigue and abrasion tests on same-w/b and strength-matched mixes; (iii) further out-of-sample comparisons of predicted and measured properties on oyster shell and on durability indicators, beyond the four strength datasets used in Section 4.4; and (iv) a plant- and supply-chain-specific life-cycle inventory.

8.5 Measurements that would most reduce the uncertainty

The Sobol indices identify where measurement would be most efficient. The life difference is led by strength-efficiency coefficient k_f (0.42), restraint factor (0.23), relative humidity (0.20) in the pavement case, air-void filling fraction (0.35), permeability factor (0.33), critical degree of saturation (0.30) in the dam case and chloride-diffusivity multiplier (0.35), surface chloride (0.17), service temperature (0.17) in the culvert case. The most informative measurements on oyster-shell concrete are therefore (i) 28-day strength at 10–40% replacement of cement at constant binder content for the actual shell and grinding, which fixes k_f; (ii) the chloride migration coefficient of the same mixes, which tests whether the filler refines the pores and therefore which row of Table 20 applies; (iii) for mass concrete, the air-void system and permeability; and (iv) a measured emission factor for the shell supply chain, because haul distance alone explains 31% of the carbon variance. Published chloride data on limestone-filler concrete could serve as proxy evidence for the transport model, but were not used here and would not substitute for measurements on shell.

9. Conclusions

- A coupled hydration–transport–damage–carbon framework with paired shell/OPC comparison, Latin hypercube Monte Carlo, Sobol sensitivity and a falsification region was developed and verified numerically. Its inert-filler strength model was rejected by published scallop-shell data (predicted 0.43 against 0.75 relative strength at 40% replacement).
- An effective strength-efficiency coefficient calibrated on one scallop-shell dataset does not transfer: on 11 independent oyster-shell points it has a mean absolute error of 0.12, no better than dilution alone (0.14), and the coefficient differs between studies from -0.40 to 1.36. The primary results therefore use a pooled distribution that expresses this heterogeneity.

- At r = 25% and constant w/b, shell replacement lowers embodied carbon by 21% and strength to 88% of OPC and shortens median life by 8.4 yr (pavement) and 5.5 yr (culvert); with strength matched the median carbon cut is −15% (P10–P90 −22 to +10%) and the life change −1.8 yr. The 80% carbon claim is reached only when strength is lost.
- The probability of a carbon cut of at least 5% with non-inferior life is at most 0.21 (pavement), 0.14 (dam, largely censoring) and 0.03 (marine culvert) at r = 25%; it is 0.00 against a 50% slag blended cement, which cuts carbon by about 42% on its own.
- These durability conclusions are conditional on unmeasured transport: with the diffusivity mapping fixed, the pavement probability at r = 25% (matched) ranges from 0.18 to 0.68, and 0.21 to 0.45 if k_f also refines the pores (Table 20). Chloride, acid and freeze–thaw transport remain uncalibrated, and the framework is a falsifiable computational screening study and not a validated service-life model; every probability is a conditional scenario result and not an empirical probability for real infrastructure.
- Leaving each study out of the pooled distribution leaves the qualitative conclusions unchanged (pavement 0.18–0.24, culvert at most 0.03) but moves the median carbon cut between −9% and −20%.
- Shell calcium carbonate used as limestone feedstock (P2) does not reduce carbon unless calcination CO_2 is credited as biogenic; durability is essentially unchanged.
- Carbon claims for shell concrete are controlled by supply chain logistics, not by material chemistry, and durability claims by the filler efficiency and the diffusivity of filler-bearing pastes; these are the quantities to measure first.
- Shell used in concrete is not available as reef substrate; the allocation of recovered shell between the two uses is treated in a companion paper (Sandler 2026b).

Declarations

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Declaration of competing interest. The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement. Leon Sandler: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work, the author used Claude (Anthropic) to help develop and test the simulation code, run the analyses and draft and revise the text and figures. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Data availability. All code, automated checks, result files and manuscript sources are openly available at https://github.com/sandlerleon/oyster-shell-concrete-mc (release v1.2.3; MIT license) and archived on Zenodo (software, https://doi.org/10.5281/zenodo.23867048; preprint of this manuscript, https://doi.org/10.5281/zenodo.23867044). The published strengths used for calibration and validation are from Abbas and Kudukkan (2025; Table 5), Kwon and Wang (2024), Kim and Lee (2023; transcription of

Abinaya et al.) and De Dios-Suárez et al. (2026), and are listed in Section 4.4.2 and in the repository. No new experimental data were collected. The companion allocation study (Sandler 2026b) is archived at https://doi.org/10.5281/zenodo.23867050.

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Classical sources without a DOI (reports, books and the Life-365 documentation) are cited from their bibliographic records; the Life-365 relation $D_{28} = 10^{(-12.06 + 2.40\ w/cm)}$ with an age exponent of 0.2 for OPC was checked against the public Life-365 documentation and secondary sources.

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Appendix A. Model parameters

All ranges are literature-style placeholders. tri(lo, mode, hi); uni(lo, hi); logn(median, σ in ln units); logu(lo, hi) is log-uniform. Parameters in the upper block are common to all cases; case-specific parameters follow.

Parameter	Key	Distribution	Unit
Hydration rate	k0	logn(0.26, 0.2)	1/d
Reaction order	nh	uni(1.5, 2.5)	-
Hydration Ea	Ea_h	uni(3.0e+4, 4.5e+4)	J/mol
Max. hydration	acap	uni(0.9, 0.97)	-
Gel strength S0	S0	logn(78, 0.1)	MPa
Strength exponent	kap_s	uni(1.8, 3)	-
Filler nucleation	nu	uni(0, 1.5)	-
Packing bonus	bpack	uni(0, 0.3)	-
Shell organics	o_org	tri(5.0e-3, 0.02, 0.04)	kg/kg
Washing efficiency	eta_w	uni(0.3, 0.9)	-
Organic retardation	kap_o	logu(5, 200)	-
Impurity penalty	kimp	uni(0, 0.05)	-
D28 factor	lamD	logn(1, 0.4)	-
D28 model form	chi_D	uni(0, 1)	-
Porosity exponent	dzeta	uni(-0.3, 0.3)	-
Age exponent m	m0	tri(0.12, 0.2, 0.3)	-
m shift	omega	uni(0, 0.3)	-
Crack–diffusion coupling	kapc	uni(2, 20)	-
Diffusion Ea	UD	uni(3.5e+4, 5.0e+4)	J/mol
Critical chloride	Ccrit	tri(0.6, 1, 1.6)	kg/m3
Propagation period	tprop	tri(3, 6, 12)	yr
Shrinkage strain	eps0	tri(400, 600, 900)	microstrain
Shrinkage time	tau_sh	tri(20, 35, 60)	d
Restraint	KR	uni(0.2, 0.6)	-
Creep relaxation	psi	uni(0.4, 0.7)	-
Early crack damage	d_cr	uni(0.03, 0.1)	-
Curling factor	f_env	uni(0.05, 0.25)	-
Air content	a_air	tri(0.02, 0.05, 0.07)	m3/m3
Critical saturation	S_crit	uni(0.8, 0.92)	-
FT damage/cycle	d1	logu(3.0e-4, 3.0e-3)	damage/cycle
FT exponent	q_ft	uni(1, 3)	-
Load variability	var_t	tri(0.85, 1, 1.15)	-
Traffic cycles/yr	n_traf	logu(5.0e+4, 1.0e+6)	cycles/yr
Failure damage	Dfail	tri(0.2, 0.3, 0.4)	-
Heat of hydration	Qcem	uni(350, 480)	kJ/kg
Thermal drop	psiT	uni(0.5, 0.9)	-
Thermal restraint	Rres	uni(0.3, 0.7)	-

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Parameter	Key	Distribution	Unit
Thermal expansion	alphaT	uni(8.0e-6, 1.2e-5)	1/K
Leaching saturation	Csat	uni(15, 22)	mol/m3
Leachable Ca	xi_L	uni(0.3, 0.7)	-
Permeability factor	kK	logu(0.03, 1)	1/m
Leaching limit	x_lim_L	tri(30, 50, 100)	mm
Ca diffusivity	Dratio_Ca	uni(0.5, 1.5)	-
Acid-accessible Ca	xiN	uni(0.4, 0.8)	-
Calcite access size	dc	logu(50, 300)	um
H+ diffusivity	DH	uni(3, 8)	-
Acid limit	x_lim_A	tri(10, 20, 30)	mm
Abrasion rate	k_ab	tri(0.05, 0.3, 1)	mm/yr
Abrasion exponent	b_ab	uni(1.5, 2.5)	-
Wear limit	w_lim	tri(15, 25, 40)	mm
Cement EF	EFcem	tri(0.75, 0.85, 0.95)	kgCO2e/kg
Raw-meal CaCO3	mCaCO3	tri(1, 1.1, 1.2)	kg/kg
Aggregate EF	EFagg	tri(4.0e-3, 7.0e-3, 0.012)	kgCO2e/kg
Water/admixture EF	EFmisc	tri(5, 10, 20)	kgCO2e/m3
Shell drying energy	e_dry	tri(20, 60, 150)	kWh/t
Grinding energy	E_ref	tri(25, 45, 90)	kWh/t
Crushing energy	E_cr	tri(5, 10, 20)	kWh/t
Electricity EF	EF_el	tri(0.1, 0.4, 0.8)	kgCO2e/kWh
Heat EF	EF_th	tri(0.2, 0.23, 0.3)	kgCO2e/kWh
Shell haul distance	dist_sh	tri(30, 150, 600)	km
Truck EF	EF_tr	tri(0.07, 0.1, 0.15)	kgCO2e/t-km
Landfill credit	c_land	tri(0, 5.0e-3, 0.02)	kgCO2e/kg
Limestone EF	EF_lq	tri(3.0e-3, 6.0e-3, 0.012)	kgCO2e/kg
Kiln penalty	kiln_pen	tri(0, 5.0e-3, 0.02)	kgCO2e/kg
Shell purity	p_sh	uni(0.92, 0.98)	-

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Case-specific parameters. Case A (pavement)/Case B (dam)/Case C (culvert):

Parameter	A	B	C	Unit
Binder content	tri(320, 350, 400)	tri(180, 230, 280)	tri(340, 370, 420)	kg/m3
w/b	tri(0.38, 0.42, 0.46)	tri(0.45, 0.5, 0.6)	tri(0.38, 0.42, 0.46)	-
Surface chloride	tri(2, 5, 9)	tri(0, 1.0e-4, 2.0e-4)	tri(3, 6, 12)	kg/m3
Cover	tri(50, 70, 100)	tri(50, 70, 100)	tri(50, 65, 90)	mm
Service temperature	tri(2, 10, 18)	tri(4, 10, 16)	tri(8, 16, 26)	C
Relative humidity	uni(55, 90)	uni(90, 100)	uni(85, 100)	%
Air-void filling	uni(0.1, 0.5)	uni(0.2, 0.7)	uni(0.3, 0.9)	-
Freeze–thaw cycles/yr	tri(10, 40, 90)	tri(0, 15, 50)	tri(0, 20, 60)	cycles/yr
Curing temperature	tri(5, 20, 35)	tri(10, 15, 25)	tri(5, 20, 35)	C
Construction emissions	tri(15, 30, 55)	tri(10, 20, 40)	tri(10, 25, 45)	kgCO2e/m3
Hydraulic gradient	–	tri(5, 20, 60)	–	-

Strength-efficiency coefficient k_f: three specifications. (i) Original model: k_f = 0 (inert filler); this generated the results of Section 7.5 labelled "original". (ii) Prior of the single-study calibration: k_f ~ U(0, 1.2), updated by rejection ABC on S0F15 and S30F15 (tolerance 0.07) to a posterior with median 0.65 (P10–P90 0.54–0.84), with the other inputs drawn from the joint posterior; this generated the results labelled "Abbas-calibrated" and is the basis of the frozen predictions of Section 4.4.2. (iii) Primary model: k_f is a common input with the pooled random-effects predictive distribution of Section 4.4.3, a Student-t with location 0.59, scale 0.82 and 3 degrees of freedom truncated to [−1, 1.5], and all other inputs at the priors of the tables above; this distribution generated every result in Sections 5–7 not labelled otherwise, the supplementary tables labelled primary, and the joint draws used by the companion paper. The study-level estimates that define it (rejection ABC with k_f ~ U(−1.5, 1.5)) are listed in Table 7.

Scenario variables: stress-ratio classes (low, medium, high) = 0.28, 0.38, 0.46 of the OPC flexural strength; acid concentration (mild, moderate, severe) = 0.03, 0.3, 3 mol H^+/m^3; chloride multiplier = 0.5, 1, 2; leaching-saturation multiplier = 0.5, 1, 1.5. Other fixed values: R = 0.1 in the S–N relation, horizon 100 yr, time step 0.125 yr, D_f ~ tri(0.2, 0.3, 0.4) (a parameter), shell and cement densities 2710 and 3150 kg/m^3.

Appendix B. Symbols and units

Symbol	Meaning	Unit
α, α_max	Hydration degree; attainable degree	–
k, k_0	Hydration rate constant (at 20 °C)	1/d
r, ρ	Shell replacement of cement (P1); of limestone (P2)	–
v_c, v_f, v_w	Volumes of cement, shell filler and water per m^3 of concrete	m^3/m^3
φ_cap, φ_p	Capillary porosity (per concrete; per paste)	m^3/m^3
X	Gel/space ratio	–
f_c, f_t, f_r, E	Compressive, tensile, flexural strength; modulus	MPa
D_{28}, D(t)	Chloride diffusivity at 28 d; at age t	m^2/s
C_s, C_crit	Surface and critical chloride content	kg/m^3
τ	Time integral of diffusivity	m^2
x, N, C_a	Front depth; capacity per volume; aggressive concentration	mm; mol/m^3; mol/m^3
D, D_m, D_f	Damage, mechanical damage, failure damage	–
S, S_eff, S_w	Stress ratio, effective stress ratio, degree of saturation	–

Symbol	Meaning	Unit
N_f	Cycles to failure	cycles
n_traf, n_ft	Traffic cycles; freeze–thaw cycles per year	1/yr
d_ft	Freeze–thaw damage per cycle	–
L	Service life (censored at 100 yr)	yr
E_gate, E_total	Cradle-to-gate; total embodied carbon	kg CO_2e/m^3
β_bio	Credited fraction of calcination CO_2	–
$\mathscr{F}$	Falsification region	–

Appendix C. Reproducing the results

```
pip install numpy scipy matplotlib
python ShellConcreteMC_v2.py --verify
python ShellConcreteMC_v2.py --reproduce --n 3000 --sobol-n 1024 --outdir out
python calibrate_abbas.py          # out-of-sample test and ABC calibration (ShellConcreteMC_v3.py)
python kf0_prediction.py           # original-model predictions for the published mixes
python make_fig_calibration.py out
python validate_independent.py          # independent oyster-shell data, per-study k_f, random-effects
distribution
python run_stage1_cal.py && python run_sobol_cal.py A 2048 && python run_sobol_cal.py B 2048 && python
run_sobol_cal.py C 2048 && python run_extras_cal.py && python run_figs_cal.py   # primary (pooled) results
python run_blend_cal.py && python run_alt_cal.py && python make_fig_validation.py shell_cal   # blended-
cement benchmark, alternative formulations, figures
```

Seeds: the calibration uses seeds 101 (draws) and 555 (k_f prior), the independent validation 2024 and 99, the primary studies seed 21, scans 31, extras 41–45, Sobol 11, the blended-cement benchmark 61, the joint draws 500–530. Environment: Python 3.11, numpy, scipy and matplotlib at the versions in requirements.txt of the repository. Studies use seed 21, severity/load scans seed 31, Sobol seed 11; the figures use fixed seeds in the script. A single full run is long on a single core; the results here were generated in stages (study, Sobol per case, figures), which the script supports through its individual functions.

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